General (for ALL areas)

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SECTION C

STATEMENT OF WORK: NATIONAL ENVIRONMENTAL SATELLITE, DATA, AND INFORMATION SERVICE (NESDIS) COMMERCIAL DATA PROGRAM SPACE-BASED ENVIRONMENTAL MONITORING (SBEM) IDIQ

C.1. INTRODUCTION

To meet the growing demand for accurate and timely environmental information, the National Oceanic and Atmospheric Administration (NOAA) is committed to maintaining a flexible, resilient, and technologically advanced observing enterprise. NOAA's strategy includes the use of both Government-owned and commercially sourced satellite data to enhance its environmental forecasting, modeling, and monitoring capabilities. To support this strategy, the National Environmental Satellite, Data, and Information Service (NESDIS) Commercial Data Program Space Based Environmental Data (SBEM) Indefinite Delivery Indefinite Quantity (IDIQ) provides a broad scope for the procurement of commercial SBEM data.

NOAA intends to acquire commercially-produced satellite-based terrestrial and space weather data across multiple Functional Categories, with a focus on Global Navigation Satellite System Radio Occultation (GNSS-RO), Global Navigation Satellite System Reflectometry (GNSS-R), Microwave Sounders (MWS), Multispectral Imagery, Infrared Sounders, Thermospheric Neutral Density measurements, and Active Sensors. These complementary technologies and measurement types provide critical observations for atmospheric profiling, environmental monitoring, numerical weather prediction, wind measurement, disaster management and space weather applications. They collectively contribute to NOAA's mission to observe and predict changes in the Earth's environment. Additionally, NOAA may consider additional SBEM data types for technologies that may, within the IDIQ timeframe, enhance NOAA's mission needs.

C.2. SCOPE

NOAA is soliciting commercial near-real-time satellite-based environmental observations. Through this acquisition, NOAA aims to operationalize and expand the use of commercial observation data to complement existing government observing systems, advance environmental monitoring, and improve weather forecasting, numerical modeling, space weather prediction, and decision support services across NOAA's mission areas. This strategy includes utilizing commercial capabilities to compliment, supplement and/or replace Government-owned and operated systems as well as integrating novel and different space-based technologies. NOAA desires procurement of commercial space-based data in seven main Functional Categories: GNSS-RO, GNSS-R, MWS, Multispectral Imagery, Infrared Sounders, Neutral Density, and Active Sensors. All data and products will be archived by NOAA. NOAA is mostly interested in raw instrument data; however, NOAA may seek value-added or derived products to meet mission objectives. In addition to the seven Functional Categories, NOAA may consider capabilities from space-based platforms to support additional environmental product areas; however, NOAA will not be evaluating additional environmental product areas during initial IDIQ awards. This IDIQ describes broader, less restrictive requirements, while Individual Task Orders within the IDIQ will stipulate more restrictive/refined/narrower requirements.

To more effectively assess and integrate novel technological advancements within the commercial sector, NOAA will utilize this IDIQ mechanism to solicit Task Orders for two purposes, Piloting and Operational Deliveries. Pilots enable NOAA to evaluate the value, cost, and exploitability of Commercial Satellite data to determine its utility, mission impact, and value to NOAA's environmental applications. Pilots will contain the necessary data delivery period but may contain a data preparation and/or a data evaluation period. Operational Deliveries are operational in nature and represent the purchase of continuous data to ingest into NOAA's operational pipeline.

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C.3. REQUIREMENTS - FOR ALL DATA TYPES

C.3.1. TEST DATA

As specified in a Task Order RFP, the Contractor shall provide:

1. A list of the satellites and their WMO identifiers that are planned to be used for data collection during the contract. Updates of future satellites should be handled per Notification requirements.

2. File name protocols, file name examples, and satellite identifier location, as well as information requested in the NCCF Interface Control Document.

As specified in a Task Order RFP, for deliveries that are not a continuation of a previous Task Order, the Contractor shall provide:

1. 24 hours of contiguous on-orbit data, representative of that to be provided under contract and compliant with the data requirements specified herein.

2. Metadata as necessary to prepare ground processing for the identified satellites.

NOAA may require the Contractor to provide limited examples of the data described in this SOW in order to validate changes to NOAA systems. For test data, the requirements for latency and delivery to the secure ingest interface may not apply.

C.3.1.1. Limited Test Data Deliveries

To meet operational data delivery requirements, NOAA may require any Contractor having new data or interfaces not yet operationally tested and validated by NOAA, to provide a 30 or more day period of test data delivery to support NOAA implementation and testing. NOAA will specify the length, data, and delivery requirements in that Task Order. NOAA may require more frequent communications with the Contractor during the Task Order.

Any Contractor determined to have a new NCCF interface shall participate in network connectivity testing with the NOAA NCCF secure ingest service and provide any supporting documentation necessary to facilitate network connectivity and data set-up.

As specified in a Task Order, the Contractor shall provide compliant data and associated metadata to enable high-fidelity NOAA system tests.

C.3.2. DELIVERY REQUIREMENTS

Minimum quantities for Task Orders will be specified within each Task Order RFP. Each Task Order RFP will describe specific data, data delivery and observation requirements, data quantities, and delivery period duration.

Measurements of all data types shall meet the following requirements:

1. The data delivery process along with file naming conventions, data formats, and metadata formats shall follow standard NESDIS methods and protocols. The Contractor shall deliver all files for secure ingest to the NESDIS Common Cloud Framework (NCCF) in accordance with the NESDIS Cloud System Data Ingest and Data Distribution Services ICD or alternative secure ingest methods such as the NESDIS Innovation Hub cloud services. The Contractor shall, upon receipt of NCCF alert notifications, resolve and, as needed, redeliver data files. Re-delivered file names shall include a date or identifier unique to that delivery.

2. The data descriptor files shall conform to the International Organization for Standardization (ISO) 19115-1:2014 and applicable amendments (content) and ISO 19139-1:2019 and applicable amendments (XML schema) standards.

3. Data shall be provided only from instruments on satellites with an assigned World Meteorological Organization (WMO) satellite identifier or a unique, internationally recognized identifier such as the COSPAR ID.

C.3.3. NOTIFICATIONS

For all Task Orders, the Contractor shall notify the Program Manager, Task Manager, COR, and the Contracting Officer, in writing, of any contract administration issues, including but not limited to changes in anticipated data delivery timelines or data specifications, as soon as they become known. For anomalies that adversely affect contractual data delivery requirements, the Contractor shall notify NOAA within 24 hours of such anomalies with an estimated time of resolution and provide a report within 30 days of the anomaly.

The Contractor shall notify the Program Manager, Task Manager, COR, and the Contracting Officer of anomalies and events that impact or have the potential to impact data deliveries or data volumes. For Operational Task Orders, the Contractor shall deliver an anomaly report to the Program Manager, Task Manager, CO, and the Contract Specialist whose framework should include these topics: 1) General information on the Task Order, Period of Performance, Contractor name; 2) Date/time of anomaly; 3) Description of the anomaly; 4) Resolution, Status, Summary, and Lessons Learned, as applicable; 5) Impact on Users, Performance, and Schedule, as applicable. The Contractor shall deliver this report within a month of the reported anomaly. For Pilot Task Orders, Contractors are encouraged, but not required, to submit an anomaly report; however, they shall report to the Government any anomaly that could adversely affect their ability to provide data during the piloted data delivery period.

For any planned platform, sensor, or ground system change that may affect data delivery or data quality, including but not limited to spacecraft, instrument, software, configuration, or other change, the Contractor shall provide written notification to NOAA of pending change no less than fourteen (14) days in advance of its implementation and shall include a description of the planned change as well as any resulting changes to the operation's concept, configuration, delivery, ground data processing, data content, or metadata. Notification is required under the contract and is not dependent upon having an active Task Order. Depending on the nature and extent of the change, the Government may request a delivery of no less than 24 hours of sample data, not to exceed 1 week of sample data, in order for the Government to implement and test any Government system modifications required to accommodate the change.

The Contractor shall document all satellite/instrument configuration changes and send notifications of such changes to the Government Task Manager. Additionally, all planned sensor and satellite constellation changes for the duration of the Task Order performance period shall be sent to the Government Task Manager.

Contractors with Operational Task Orders shall deliver information reports on the health of the Contractor’s ground system architecture, on-orbit sensors, and constellation status. Whenever systems fail to function as expected, the Offeror must provide a detailed description of any anomalies, including the estimated time for repair of the affected architecture (ground and/or on-orbit). At the time of reporting, the Contractor shall provide an operating status and capacity of their payload for each spacecraft. Contractors shall provide this information no later than one week after the end of each 3-month reporting period, as defined in each Task Order RFP. The Government may specify more frequent reporting requirements for Pilots.

Contractors with Operational Task Orders shall submit a 12-month forecast detailing projected additions and losses to their on-orbit capabilities, along with a breakdown of the first three months, including scheduled maintenance and anticipated changes to their operational capabilities. The Contractor shall deliver these information reports no later than one week after the end of each 3-month reporting period, as defined in each Task Order RFP.

C.3.4. GENERAL REQUIREMENTS

The Contractor shall be available to discuss their data, support data integration, discuss anomalies, support technical exchanges, and provide sustaining engineering information and support through the end of the contract Ordering Period.

The Contractor shall support mitigation of NOAA Supply Chain Risk Assessment findings as determined by the NOAA OCIO.For operational task orders, the Contractor shall provide a point of contact that is available 24x7 and will respond within 60 minutes of notification by NOAA to investigate and resolve any data delivery or performance issues.

Unless specified in a Task Order, when the Contractor has an existing agreement to provide a specific type of data with a latency equal to or less than the latency required by this contract, and the terms of that agreement permit immediate distribution and use by NOAA to other U.S. Government agencies, National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, or members of the Coordination Group for Meteorological Satellites such as the European Organisation for the Exploitation of Meteorological Satellites (EUMETSAT), then the Contractor shall not provide the same type of data to NOAA to fulfill the delivery requirements of this contract and its Task Orders.

C.3.5. CONTRACTING OFFICER’S REPRESENTATIVE AND TASK MANAGER

In accordance with CAR Clause 1352.201-72, a Contracting Officer's Representative (COR) will be designated at award. The COR may provide technical direction and has the authority to accept delivery. The COR is not authorized to make any commitments or otherwise obligate the Government or authorize any changes which affect the contract price, terms, or conditions of

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the award. Any Contractor request for changes shall be referred to the Contracting Officer directly or through the COR. No such changes shall be made without the express written prior authorization of the Contracting Officer.

A Task Manager (TM) will be designated at award. The TM manages the contract cost, schedule, performance, and risks. The TM does not have delegated authority to represent the Contracting Officer or bind the Government. The TM cannot change, waive, add, or request any additional specifications or performance.

C.3.6. SECURITY REQUIREMENTS

The Contractor shall comply with CAR 1352.239-72, section (i). The Contractor may request COR approval of alternate security accreditation methodologies for compliance with section (i) including: Information security assessment methodology promulgated by the U.S. Federal Government (specifically, the latest revisions of National Institute of Standards and Technology (NIST) Special Publication (SP) 800-37, NIST SP 800-53, and NIST SP 800-171), or internationally recognized private industry information technology (IT) security frameworks (such as International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) 27033, 27001, 27006 and 27002, or Control Objectives for Information and Related Technology (COBIT)). The Contractor will provide an analysis for alternative methodologies compared to NIST SP 800-53 high impact baseline identifying any gaps.

The Contractor shall deliver files to NOAA ingest using either Hypertext Transfer Protocol Secure (HTTPS) (preferred) or Secure File Transfer Protocol (SFTP). The Contractor shall use a data integrity method compliant with the NESDIS Cloud System Data Ingest and Data Distribution Services Interface Control Document (ICD) and approved by NOAA.

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C.3.7. DATA RIGHTS

Per this contract, the Contractor grants NOAA a perpetual, non-exclusive, irrevocable, worldwide license to use, reproduce, and create derived and value-added products with the Contractor-provided data in any manner and for any purpose and to authorize others to do so on its behalf (including other contractors and recipients of financial assistance awards).

Copyright Assignment: By performing work under this contract and providing deliverables, the Contractor assigns NOAA ownership (including copyright, as applicable) of all contract deliverables, including, but not limited to, raw and processed data and documentation, for all work performed under this contract.

A ***derived product*** is defined as a work that is created when data is manipulated to such a degree that it cannot be reverse engineered. Derived products created using

Contractor-provided data received under this contract are not subject to the data rights provisions in this section or any other contractual, licensing, or intellectual property claims by the Contractor, and therefore may be publicly distributed without use restrictions.

A *value-added product* is defined as a work that is created when data is modified through technical manipulation, addition of data, or both, where the principal features and characteristics of the source data are retained in the work and are extractable through technical means. Value-Added Products created using Contractor-provided data received under this contract are subject to the data rights provisions in this section.

Level 1 and Level 2 products created by NOAA or its affiliates from Contractor-provided data are considered to be *value-added products*. All other products created by NOAA or its affiliates from Contractor-supplied data, such as model outputs, are considered to be *derived products*.

Furthermore, the Contractor shall provide pricing for various external data sharing rights, including but not limited to the external distribution and use rights described in the options below. The Government will list specific data sharing rights in each Task Order RFP.

Option 1: Right to distribute all Contractor-provided data to any entity immediately after receipt at NOAA with no restrictions on their use or further distribution, under a Creative Commons Attribution 4.0 International license (CC BY 4.0).

Option 2: Right to distribute all Contractor-provided data to U.S. Government agencies, International Partners, including but not limited to those responsible for national meteorological, space weather, or hydrological services, and National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, members of the Coordination Group for Meteorological Satellites, and non-profit entities and academic entities, immediately after receipt at NOAA, for non-commercial use but not for further distribution.

Option 2A: In addition to the rights under Option 2, Option 1 rights will apply to all Contractor-provided data beginning at 24 hours after collection.

Option 3: Right to distribute all Contractor-provided data to U.S. Government agencies, International Partners, including but not limited to those responsible for national meteorological, space weather, or hydrological services, and National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, and members of the Coordination Group for Meteorological Satellites, immediately after receipt at NOAA, for non-commercial use but not for further distribution.

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Option 3A: In addition to the rights under Option 3, Option 1 rights will apply to all Contractor-provided data beginning at 24 hours after collection.

Additional Data Sharing Options: The Government may consider alternatives, to include more restrictive, data sharing terms and conditions. Contractors shall describe any alternative data sharing provisions as desired.

NOAA will ensure that any non-NOAA entity receiving restricted data, including documentation, is made aware of the identified use and distribution restrictions associated with the data sharing rights specified in the IDIQ contract and initial and subsequent Task Orders. If NOAA becomes aware that an entity is not adhering to those restrictions, NOAA will seek to ensure compliance, and if unsuccessful, may terminate the entity's access to the data. NOAA, however, has no additional obligations under the IDIQ contract or initial and subsequent Task Orders related to data use by third parties. In no event is NOAA liable to the contractor for third party use or release of contractor data.

C.3.7.1. Documentation

In addition to data deliveries, the Government requests specific documentation to support understanding, usage, and validation of the data required for each Functional Category. Contractors shall define the distribution rights for the documentation required, as listed in each Functional Category. All documentation provided under each Functional Category that is marked as 'Proprietary' or 'Confidential'—including detailed instrument specifications and satellite engineering data—shall be treated as vendor-proprietary. NOAA agrees to protect such documentation from unauthorized disclosure and will not share, publish, or distribute it to non-NOAA entities or third parties without the express written consent of the Contractor.

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C.4. GNSS-RADIO OCCULTATION (RO) DATA IDIQ REQUIREMENTS

C.5. GNSS-REFLECTOMETRY DATA IDIQ REQUIREMENTS

C.6. MICROWAVE SOUNDER DATA IDIQ REQUIREMENTS

C.7. MULTISPECTRAL VISIBLE/INFRARED IMAGERY DATA IDIQ REQUIREMENTS

C.8. INFRARED SOUNDER DATA IDIQ REQUIREMENTS

C.9. THERMOSPHERIC NEUTRAL DENSITY DATA IDIQ REQUIREMENTS

C.10. ACTIVE SENSORS DATA IDIQ REQUIREMENTS

C.11. ADDITIONAL SBEM DATA IDIQ REQUIREMENTS

(End of Section C)

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FC1: GNSS-RO Requirements

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C.4. Functional Category 1: GNSS-RADIO OCCULTATION (RO) DATA IDIQ REQUIREMENTS

C.4.1. INTRODUCTION

In order to respond to the demand for environmental information, NOAA strives for an observing enterprise that is flexible, responsive to evolving technologies and economically sustainable, while continuing to be a leader and member of the global meteorological community. To help meet these goals and satisfy observational requirements, NOAA uses satellite-based global navigation satellite system (GNSS) radio occultation (RO) data purchased from multiple commercial contractors to complement Government sponsored sources for use in weather models and other systems. GNSS-RO data are valued for their high accuracy, long-term stability, and lack of onboard calibration needs, making them particularly well-suited for assimilation into numerical weather prediction (NWP) models and space weather applications. NOAA is interested in continuing the purchase, ingest and exploration of additional capabilities within these RO data observations.

C.4.2. SCOPE

NOAA is soliciting commercial near-real-time satellite-based GNSS-RO measurements (including polarimetric RO (PRO) and ionospheric specifications) that will be processed into neutral atmosphere and space weather products. These derived products will be fed into NOAA's operational data systems, including weather and space weather analysis and prediction systems, and used for weather, climate, and atmospheric research purposes. All data and products will be archived by NOAA.

C.4.3. DATA REQUIREMENTS

Contractors must have the capability to fulfill Task Orders for data meeting the requirements specified in Section C.3. Optionally, contractors may propose the capability to fulfill Task Orders for data meeting requirements specified in Section C.4.

C.4.3.1. Definitions

The following definitions are provided in the context of all GNSS-RO measurements:

1. An RO sounding is a time series of satellite-based RO open loop measurements made during a GNSS occultation event from which NOAA can produce a quasi-vertical series of bending angles.

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2. A PRO sounding is composed of two concurrently measured, consistently time-tagged RO soundings, one tracked with horizontal linear polarization, and one tracked with vertical linear polarization. Polarization orientation is defined with respect to the spacecraft local level.
3. An ionospheric track is a time series of satellite-based GNSS dual-frequency pseudorange, carrier phase, and signal-to-noise ratio (SNR) measurements from which NOAA can produce geolocated ionospheric products for space weather applications and research purposes. Ionospheric information obtained from ionospheric tracks include the following:
    - a Total Electron Content (TEC): Total number of electrons present along the GNSS's signal path between the GNSS transmitter and RO receiver. Electron density profiles are derived from TEC data.
    - b Scintillation Amplitude, S_4: Characterizes the variations in the SNR of the signal.
    - c Scintillation Phase, $\sigma\varphi$ (sigma phi): Quantifies the severity of rapid fluctuations in the GNSS signal phase.
4. The elevation angle of an observation is defined as the angle between the spacecraft local level and the ray from the spacecraft receiver antenna to the GNSS transmitter antenna.
5. The height of straight line (HSL) altitude of an observation with negative elevation angle is the height of the tangent point of a straight line between the GNSS transmitter and the receiver relative to the World Geodetic System 1984 (WGS84) reference ellipsoid.
6. The time of an RO/PRO sounding is defined as the GPS time of the zero HSL altitude point.
7. The time of an ionospheric track is defined by the GPS time of the first observation making up the track.
8. The latency of a data product is defined as the difference between the time of the observation and receipt of the data into the NESDIS Common Cloud Framework (NCCF).
9. The daily count of a data product is defined as the number of RO soundings or ionospheric tracks in a 24 hour period starting at 00:00:00 GPS time.
10. The location of an RO sounding is defined as the geographic location of the zero HSL altitude point; the location of an ionospheric track is defined as the geographic location of the 120 km HSL altitude point.
11. The latency of an ionospheric observation is defined as the difference between the time of the observation epoch and the receipt of the observation data by the NESDIS

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Common Cloud Framework (NCCF). The daily median latency for ionospheric data is computed from all epochs collected during the 24 hour period beginning at 00:00:00 GPS time.

12. The ionospheric observation epochs within an ionospheric track are defined as the individual measurement times within that track.
13. The Carrier Phase Sampling Rate is the underlying rate of both carrier phase and SNR observation collection used in the on-board calculation of scintillation indices such as S4 and sigma-phi. The Scintillation Index Calculation Time Interval is the time period over which a single index calculation is made.
14. Level 0 data: Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed.
15. Level 1a data: Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information.

C.4.3.2. Specific Geospatial Distributions

The required geographic distribution of RO soundings or ionospheric tracks may be specified in each Task Order RFP. As specified in a Task Order, the Government may require the Contractor to provide one or more of the following:

1. For polar orbits only, the number of RO soundings or ionospheric tracks provided each day shall not vary by more than a factor of four (4) between any 30-degree latitude band.
2. For all orbits, within any 30-degree latitude band, the number of RO soundings or ionospheric tracks provided each day within a 30 degree longitude section shall not vary by more than a factor of four (4) from any other 30 degree longitude section.
3. Provide increased quantities of data within specific geographical regions as defined in a Task Order RFP.

C.4.3.3. Neutral Atmosphere Radio Occultation Data Requirements (Category 1)

Data from RO soundings will be used to produce bending angle profiles of the neutral atmosphere to obtain specification of refractivity, temperature, pressure and moisture. Contractors proposing to deliver neutral atmosphere data must have the capabilities to provide data meeting the following requirements:

1. RO soundings shall be collected as continuous dual-frequency open-loop measurements at either 50 Hz or 100 Hz.
2. Data shall be provided only from instruments on satellites with an assigned World Meteorological Organization (WMO) satellite identifier.
3. If applicable, the signal-to-noise ratio (SNR) conversion factor from dB-Hz to Volts/Volt (V/V) in a 1 Hz band shall be provided.
4. The SNR of the stronger signal in a dual frequency signal pair, when averaged over HSL altitudes of 60 km to 80 km, shall be greater than 200 V/V in a 1 Hz band.
5. The HSL altitudes of occultations shall span at least -150 km to 90 km.
6. RO soundings shall be from GPS, GLONASS, Galileo, or BeiDou GNSS satellites.
7. RO soundings shall be provided with concurrent precise orbit determination (POD) tracking data.
8. RO soundings shall be provided with concurrent POD tracking data such that one or two POD tracks may serve as reference links for single difference processing of a RO sounding.
9. POD tracking data shall include dual-frequency pseudo range, carrier phase, and SNR, as collected on-orbit in real time.
10. POD tracking data shall include dual-frequency measurements recorded at 1 Hz or greater rate from a minimum of four GNSS satellites simultaneously for 95% of the POD data collection period.
11. Every other orbit of each RO satellite from which data are provided shall be covered by POD tracking data at no less than 50% duty cycle with continuous arc length of at least 50 minutes per orbit.
12. POD tracking data shall be from GPS satellites, plus optionally Galileo and/or GLONASS satellites.
13. Navigation solutions calculated onboard the spacecraft shall be provided, including receiver clock offsets and time-tagged Earth-Centered-Earth-Fixed position and velocity.
14. Navigation solution data shall be provided at intervals of 30 seconds or less.
15. RO satellite attitude quaternions shall be collected concurrently with all POD and RO data at intervals of 30 seconds or less.
16. All RO soundings shall have a latency of 140 minutes or less.

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17. The time of RO soundings shall be nearly uniformly distributed in time of day such that the quantity delivered in any six-hour period of the UTC day, when averaged over 30 consecutive days, shall vary by less than a factor of two from any other six-hour period.

18. At a minimum, the Contractor must have the capability to provide a daily count of at least 500 RO soundings, when averaged over any 30-day period, and with no more than three days in any 30 day period falling beneath 400/day. 500 RO Soundings/day represents the minimum ordering for Task Orders. The Government will stipulate required quantities in future Task Order RFPs.

19. Unless otherwise specified in a Task Order, the daily count of compliant RO soundings delivered by the Contractor, when averaged over any continuous fifteen-day period, shall meet or exceed the required target count as specified in the Task Order contract, and shall not fall below 80% of the required target count for more than three days in any thirty day period.

20. Metadata shall include, at a minimum, for each satellite/instrument configuration:
    1. Satellite mass, center of mass, external dimensions including any protruding appendages (e.g., solar array wings, antennae), satellite coordinate system geometry, POD and RO antenna reference point vectors relative to the center of mass and corresponding uncertainty estimates, antenna boresight vectors, and antenna fields of view.
    2. RO/PRO and POD antenna phase center offset and phase center variation measurements and corresponding uncertainty estimates, in ANTEX format (https://files.igs.org/pub/data/format/antex14.txt).

C.4.3.4. Polarimetric Radio Occultation (PRO) Data Requirements (Category 2)

Data from PRO soundings will be used to produce bending angle profiles of the neutral atmosphere to obtain information about hydrometeors. Additionally, PRO soundings will be combined to produce bending angle profiles of the neutral atmosphere to obtain specification of refractivity, temperature, pressure and moisture. Contractors proposing to deliver PRO data must have the capabilities to provide data meeting the following requirements. The Government may stipulate, in a Task Order RFP, additional detail concerning the specifics of these thresholds. The Government will stipulate required quantities in future Task Order RFPs.

1. Meet all the data requirements as specified in Section C.4.3.3, Neutral Atmosphere Radio Occultation Data Requirements (Category 1) for each polarization direction (horizontal (H) and vertical (V)).

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2. PRO soundings shall be collected from separate orthogonal linear horizontal (H) and vertical (V) polarizations with respect to the spacecraft local level.

3. Time-tags for the horizontal and vertical polarization measurements shall be consistent and refer to the same frequency reference.

4. The horizontal and vertical polarization measurements shall be able to be independently processed into retrieved profiles, and be able to be combined into a single circularly-polarized profile.

5. Antenna pattern information, as defined in Section C.4.3.2., shall be supplied for both horizontal and vertical polarization components.

C.4.3.5. Space Weather Data Requirements (Category 3)

Ionospheric measurements will be used to produce space weather products including total electron content (TEC) and scintillation indices (scintillation amplitude S_4, and scintillation phase, $\sigma\varphi$). Contractors proposing to deliver space weather data must have the capabilities to provide Level 0 data meeting the following requirements. The Government may stipulate, in a Task Order RFP, additional detail concerning the specifics of these thresholds.

C.4.3.5.1 Requirements for Total Electron Content (TEC)

1. TEC Track extent: Tracks must be at least 8 minutes in duration and span from 90 km HSL to the maximum possible elevation angle for the transmitter/receiver geometry, which must occur at a positive elevation angle.

2. Pseudorange, carrier phase sampling rate: 1 Hz

3. Absolute TEC Measurement range: 0 to 2000 TECU

4. Maximum measurement uncertainty:

    a. Relative 0.5 TECU RMS

    b. Absolute 4 TECU RMS

5. Maximum daily median latency: 30 minutes

6. Deliver all required observation data needed for each TEC track from each participating satellite to the NOAA NESDIS Common Cloud Framework (NCCF) in time-ordered data batches.

7. Ensure that TEC ionospheric tracks include a minimum of 100,000 dual-frequency (1 Hz) observation pairs (i.e., from two transmitters) per day per spacecraft per constellation above 20° local elevation.

8. At a minimum, include metadata detailing spacecraft ID, observation time, receiver information, satellite position, raw measurement uncertainties, and data quality flags in the data delivered.
9. Provide sustained coverage of ionospheric tracks across various latitudes and longitudes.
10. At a minimum, the Contractor shall provide data meeting these TEC requirements from at least 500 ionospheric tracks per day, when averaged over any 30-day period, and with no more than three days in any 30 day period falling beneath 400/day. The Government will stipulate required quantities in future Task Order RFPs.
11. Unless otherwise specified in a Task Order, the daily count of compliant ionospheric tracks delivered by the Contractor, when averaged over any thirty day period, shall meet or exceed the required target count, and shall not fall below 80% of the required target count for more than three days in any thirty day period.

C.4.3.5.2. Requirements for Scintillation Indices & Associated High Rate Observations

1. Scintillation amplitude indices, S_4, shall be computed on-orbit based on high rate SNR observations and downloaded for all tracks. If feasible, scintillation phase indices, $\sigma\varphi$ (Sigma phi), shall be computed on-orbit based on high-rate carrier phase observations and downloaded for all tracks. The underlying high rate (HR) SNR and carrier phase observations associated with tracks with on-board S_4 or $\sigma\varphi$ that exceed certain thresholds shall be included in the sensor telemetry. The Government may stipulate, in a Task Order RFP, additional detail concerning the specifics of these thresholds and other related parameters. The government may also stipulate specific detrending algorithms to be used in the $\sigma\varphi$ calculations. Additionally, the Contractor shall have the ability to deliver all scintillation data for a period of not less than 20 seconds before and after scintillation indices meet a predetermined specification threshold, as defined in a Task Order RFP.
2. Minimum SNR/carrier phase sample rate: 50 Hz
3. High rate data time interval used for on-board S_4 and $\sigma\varphi$ calculations: 10 s
4. On-board S_4 and $\sigma\varphi$ calculation cadence: 10 s
5. S_4 Measurement range: 0 to 1.5
6. $\sigma\varphi$ (Sigma phi) Measurement range: 0.1 to 20 rad
7. S_4 Precision: $\leq$ 0.1

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8. Maximum daily median latency: 30 minutes
9. The Contractor shall have the ability to provide high rate scintillation observations and indices in pre-defined latitudinal ranges, as specified in each Task Order RFP.

C.4.4. ENHANCED DATA REQUIREMENTS

C.4.4.1. Neutral Atmosphere Data with Enhanced Signal-To-Noise Ratio

Task Orders may be issued which require higher SNR data. When required by NOAA, the SNR of the stronger signal in a dual frequency signal pair, when averaged over HSL altitudes of 60 km to 80 km, shall be greater than 1200 V/V in a 1 Hz band.

C.4.4.2. Temporal Resolution Enhancements

NOAA may include specific requirements on the temporal distribution of RO soundings or ionospheric tracks, to include different Sun-Synchronous Orbit (SSO) local times and/or lower-inclination orbits (non-SSO). When required in a Task Order, the local time of the delivered RO soundings or ionospheric tracks shall be within the specified local time range or group of ranges. The extent of any local time ranges specified in a Task Order will be at least 1 hour.

C.4.4.3. Designated Satellites, Instruments or Modes

NOAA may limit the data in a Task Order to come from a specific predesignated list of satellites, instruments, or modes (for example, rising vs. setting occultations). Such a restriction would allow NOAA to coordinate data purchases with other Agencies and partners to avoid duplication of shared observational data.

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C.4.5. DELIVERY REQUIREMENTS

C.4.5.1. Data Quantity and Delivery Duration

The daily data quantity, volume and delivery period duration will be specified in each Task Order RFP.

The processing level of the RO soundings or ionospheric tracks and associated data to be delivered, i.e. either Level 0 or Level 1a, may be further specified in a Task Order RFP.

C.4.5.2. Data Formats

1. The Contractor shall deliver data as Level 0 files or both Level 0 and Level 1a files, as specified by each Task Order. If Level 1a, the contractor shall deliver RO observations and

high rate observations for derivation of scintillation products in opnGns format (https://cdaacwww.cosmic.ucar.edu/cdaac/cgi_bin/fileFormats.cgi?type=opnGns).

   a. For PRO data, the Level 0 files shall include separate data for each polarization, horizontal and vertical. If PRO data are supplied in opnGns format, horizontal and vertical polarization data shall be supplied as separate files by antenna ID, with one antenna ID per physical antenna and per polarization.

2. The Contractor shall deliver POD tracking data as Level 0 files or both Level 0 and Level 1a files, as specified in each Task Order. If Level 1a, the contractor shall deliver POD tracking data in RINEX2.20 or above format (https://files.igs.org/pub/data/format/).

3. The Contractor shall deliver on-orbit computed navigation solution data as Level 0 files or both Level 0 and Level 1a files as specified in each Task Order. If Level 1a, the contractor shall deliver navigation data in SP3d format (https://files.igs.org/pub/data/format/sp3d.pdf).

4. The Contractor shall deliver attitude quaternion data as Level 0 files or both Level 0 and Level 1a files, as specified in each Task Order. If Level 1a, the contractor shall deliver Level 1a attitude data in leoAtt format (https://cdaac-www.cosmic.ucar.edu/cdaac/cgi_bin/fileFormats.cgi?type=leoAtt).

5. The Contractor shall deliver a detailed Level 0 data decoding dictionary for each to-be delivered Level 0 data type. If delivered in binary format, in addition to the dictionary, the Contractor shall deliver binary decoder software as source code.

6. The data descriptor files shall conform to the International Organization for Standardization (ISO) 19115-1:2014 (content) and ISO 19139 (XML schema) standards, in accordance with the NOAA Environmental Data Management Framework.

7. Data files shall be named according to the file naming conventions used by the University Corporation for Atmospheric Research (UCAR) COSMIC Data Analysis and Archive Center (CDAAC), as defined at https://cdaacwww.cosmic.ucar.edu/cdaac/doc/formats.html.

8. The Contractor shall deliver data to NOAA batched as compressed TAR files, with each

   TAR file including concurrent POD and attitude data files. RO data files may be included in the POD and attitude TAR files or delivered as separate TAR files with file names correlating file types.

9. When RO data at both Level 0 and Level 1a are to be delivered, the Contractor shall deliver Level 0 and Level 1a data in separate TAR files.

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10. Alternative File formats: Offerors who cannot comply with the file name convention used by the CDAAC shall propose an alternative naming convention. Acceptance of any alternative naming conventions is at the discretion of the Government. Additionally, the Government may request alternative formats from Offerors, as specified in operational Task Order RFPs.

C.4.5.3. Data Delivery and Documentation

1. The Contractor shall, upon receipt of NCCF alert notification, resolve and (as needed) redeliver data files when data latency requirements can still be met. Re-delivered file names shall include a date or identifier unique to that delivery.

2. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. Technician documentation shall include, at a minimum, the following:

    a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.

    b. Documentation to support quality control and calibration/validation activities.

    c. Antenna pattern designs.

    d. User guides and detailed data format and data specification documents.

    e. Description of instrument(s) used in the data acquisition and their sampling characteristics.

    f. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

C.4.5.4. GNSS-RO Test Data

In addition to specifications listed in Section C.3.1, within 7 days of contract award, the Contractor shall provide:

1. A description of the onboard receiver clock steering methods, sufficient to enable NOAA to determine any needed processing adjustments on the ground, shall be provided.
2. If the POD and RO receiver clocks are not the same, the Contractor shall provide a description of how to align them for single difference RO data processing.

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FC2: GNSS-R Requirements

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C.5. Functional Category 2: GNSS-REFLECTOMETRY DATA IDIQ REQUIREMENTS

C.5.1. INTRODUCTION

Global Navigation Satellite System Reflectometry (GNSS-R) is a remote sensing technique that utilizes signals from positioning, navigation, and timing (PNT) satellites, such as GPS and Galileo, to measure and study properties of the Earth's surface. NOAA has previously performed significant scientific work on investigating the utility of both government-owned and commercially sourced GNSS-R data for use in numerical weather prediction (NWP), deriving ocean surface wind (OSW) speeds, characterizing sea ice, measuring soil moisture, and monitoring flooding and inundation. GNSS-R offers a cost-effective approach for frequent and wide-area monitoring of the ocean surface and other surface properties, thus enhancing marine weather forecasting, tropical cyclone monitoring, and hydrological applications.

The use of GNSS-R for OSW speed retrievals has the potential to allow for regular monitoring of surface winds over large areas at a relatively low cost. GNSS-R observations can generate wind products for tropical cyclone and winter storm prediction and marine weather forecasting. By leveraging GNSS-R wind speed measurements with other technologies, researchers and scientists can contribute to improving weather forecasting and tropical cyclone prediction. NOAA desires scalable and operationally viable wind retrieval at the sea/atmosphere interface in all weather conditions via GNSS-R systems.

GNSS-R observations support understanding of hydrological processes such as evaporation and precipitation, which influence the global energy and moisture cycle. These observations provide actionable intelligence for sea-ice characterization, soil moisture estimation, flood and inundation monitoring. This data can effectively support activities including disaster management and assimilation into NWP models for improved forecast skill in areas such as convective initiation.

C.5.2. SCOPE

NOAA is soliciting commercial, near-real-time satellite-based GNSS-R observation data to support operational forecasting and environmental monitoring. NOAA will process these measurements into geophysical products, assimilate them into numerical weather prediction (NWP) models and downstream applications, and archive both the raw data and the derived datasets for long-term access and scientific research.

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C.5.3. DATA REQUIREMENTS

All delivered data, including associated metadata and ancillary files, shall conform to NOAA data standards or be fully compatible with NOAA operational data systems, formats, and protocols, to ensure seamless integration, processing, and archival within NOAA's enterprise systems.

C.5.3.1. Definitions

The following definitions are provided in the context of all GNSS-R measurements:

1. **Antenna Gain**: The measure of how well an antenna directs radio frequency energy in a specific direction, affecting both transmitted and received signal strength.
2. **Geophysical Model Function (GMF):** OSW measurement by GNSS-R, SAR, and Scatterometry works by relating the Radar cross section to sea surface roughness, which is primarily determined by the local wind speed. This measurement is facilitated by a Geophysical Model Function, which is an empirical relationship linking NBRCS, incidence angle, wind direction, and wind speed.
3. **GNSS-R:** A passive bistatic remote sensing technique that leverages existing signals from positioning, navigation, and timing (PNT) satellite constellations such as GPS, Galileo, GLONASS, and BeiDou. In this approach, GNSS signals are reflected off the Earth's surface and received by spaceborne GNSS-R receivers. As GNSS signals interact with the sea surface, wind-driven wave fields modify the signal's amplitude, phase, Doppler spread, and polarization. These variations are captured in GNSS-R system Level 1A Delay-Doppler Maps (DDMs) and further processed into Level 1B calibrated bistatic radar cross-section (σ^0) values. From these calibrated reflections, the user can derive Level 2 geophysical observables, including wind speed-sensitive reflectivity and coherence indices.
4. **GNSS-R Delay-Doppler Map (DDM)**: A two-dimensional representation of reflected GNSS signal power as a function of time delay and Doppler frequency, used to analyze surface scattering properties.
5. **GNSS-R Specular Point**: The location on the Earth's surface where the angle of incidence of the GNSS signal equals the angle of reflection toward the receiver, representing the primary reflection point.
6. **GNSS-R Track**: The ground-projected path traced by the specular reflection point as the GNSS signal reflects off the Earth's surface and is observed by the receiver over time.

7. **Latency**: The time interval between the observation of the data (e.g. GNSS-R reflection event) and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).

8. **Level 0**: Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed. For GNSS-R, this includes raw intermediate frequency (IF) samples or complex baseband representations without calibration, geolocation, or additional metadata. Aligns with NOAA OSW Level 0 observations.

9. **Level 1**: Instrument-processed, calibrated data at full resolution, time-referenced, and annotated with ancillary information. For GNSS-R, this includes DDMs and derived σ^0 values with associated metadata and basic geolocation. Aligns with NOAA OSW Level 1A (DDMs) and Level 1B (σ^0) observations.

10. **Level 2**: Geophysical retrieval-ready observables derived from calibrated Level 1 data. For GNSS-R, this includes normalized reflectivity, coherence index, and effective specular area. These observables shall be suitable for NOAA OSW environmental retrievals for scientific and operational use.

11. **Normalized Bistatic Radar Cross-Section (NBRCS, σ^0)**: A normalized measure of the strength of the scattered GNSS signal from the surface, accounting for geometry and antenna gain, used for geophysical retrievals.

C.5.3.2 GNSS-R Data Requirements

The requirements set forth below constitute data required to meet the following specific operational measurement requirements for the GNSS-R technology. The Government may further specify Level 0, Level 1 and Level 2 requirements in a future Task Order RFP.

1. GNSS-R observations shall be delivered at the following levels:

   a. Level 0 data including:
      i. Full resolution DDMs.
      ii. Compressed DDMs with a reduced range of delay and Doppler.

   b. Level 1 A and B products including:
      i. Calibrated DDMs. Observations will include all data necessary to achieve fully calibrated DDMs and normalized bistatic radar cross sections. Data must be radiometrically calibrated to correct for system biases, antenna patterns, and platform effects. DDMs may be calibrated to Watts or calibrated using the ratio of direct to reflected received power (Watts/Watt).

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ii. Specular reflection point location estimates.

c. Provide the following related data for use in NOAA's analysis team:
   i. Quality flags generated in processing pertaining to all levels of provided data and relevant documentation. Quality flags will include signal quality, land/ocean discrimination, ice flags, and navigation or receiver errors.
   ii. All data used for calibration/validation and product development.
   iii. Data readers and format descriptions (i.e., reader software source, data dictionaries, etc.) necessary for processing and analysis.

C.5.4. OBSERVATION REQUIREMENTS

1. All GNSS-R observations shall meet or exceed the parameters summarized in Table 1. Each Task Order request for proposal (RFP) will contain more specific observation requirements to include, but not limited to, frequency band coverage, swath, resolution, accuracy, latency, refresh and coverage.

Table 1. GNSS-R Observation Requirements

Performance Level	Threshold Requirement
Horizontal Resolution	25 km
Accuracy (Speed)	2 m/s or 10% of the measurement
Latency	180 minutes

2. GNSS-R data observations shall meet or exceed the following quality and quantity requirements as described in this section. Each Task Order RFP may contain more specific observation requirements.

   a. Unless otherwise specified in a Task Order RFP, near real-time data with data received less than 3 hours after observation. Continuous data collection with all recorded data provided.
   b. Data delivery of Cyclone Global Navigation Satellite System (CYGNSS)-class peak gain (> 10 dBi) GNSS-R observations and/or data delivery of high gain (> 18dBi) GNSS-R observations. Specific gain requirements for acquired data will be specified in Task Order RFPs.
   c. Data delivery shall consist of complaint tracks. A compliant track is generally defined by a set of Quality Control (QC) flags and is typically considered compliant if it satisfies conditions including:

i. Signal-to-Noise Ratio (SNR) threshold: The peak of the Delay-Doppler Map (DDM) must be sufficiently above the noise floor. For example, a SNR of < 2dB may be considered not compliant.

ii. Geometric Validity: The "specular point" (the point of reflection) must be accurately geolocated. For example, if the geometry is too "grazing" with low elevation angles, the signal may be discarded due to tropospheric delay errors.

iii. Antenna Gain: The reflection must occur within the main beam of the receiver's antenna. For example, if the specular point falls in a "side lobe" or at the very edge of the antenna pattern, the power calibration may be unreliable.

iv. Track Continuity: The receiver must maintain a consistent "lock" on the signal's code phase and Doppler frequency. For example, if the receiver skips or loses the track, the resulting fragmented data may be non-compliant.

The standards for compliant tracks and QC flags will vary by application and the precise definition of compliant GNSS-R tracks will be specified in each Task Order RFP.

d. Unless otherwise specified in a Task Order, the Contractor shall have the capability to provide no less than 200 tracks per day of compliant near-nadir GNSS-R observations. 200 GNSS-R tracks/day represents the minimum ordering for Task Orders.

e. Tasking ability of satellite(s) to provide higher quantities of system tracks per day during tropical cyclone events, as specified in a Task Order RFP.

f. Unless otherwise specified in a Task Order RFP, the daily count of compliant GNSS-R tracks delivered by the Contractor, when averaged over any continuous thirty-day period, shall meet or exceed the required target count as specified in the Task Order contract, and shall not fall below 80% of the required target count for more than three days in any thirty day period.

g. Separation of ocean, land, and ice data into separate data streams to make efficient access for separate applications.

h. Spatial resolution of 25 km x 25 km or better for GNSS-R OSW determination as specified in each Task Order RFP.

i. Geographic distribution of GNSS-R tracks shall be worldwide or as specified in each Task Order RFP.

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C.5.5. DELIVERY REQUIREMENTS

All GNSS-R measurements shall meet the following requirements as described in this section. Each Task Order RFP may contain more specific delivery requirements.

1. Metadata shall include, at a minimum, for each satellite/instrument configuration:

    a. Satellite mass, satellite coordinate system geometry, center of mass location, external dimensions including any protruding appendages (e.g., solar array, antennae) and, if appropriate, their variation with time (e.g. articulating solar panel).
    b. Antenna phase center offset and phase center variation measurements and corresponding uncertainty estimates representing the as installed on-orbit configuration for all relevant frequencies, in ANTEX format (https://files.igs.org/pub/station/general/antex14.txt).
    c. Antenna gain patterns representing the as installed on-orbit configuration for all relevant frequencies, and file format documentation.
    d. Measurements of sensor bias and/or drift, if appropriate.

2. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. Technical documentation shall include, at a minimum, the following:

    a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.
    b. Documentation to support quality control and calibration/validation activities.
    c. Antenna pattern designs.
    d. User guides and detailed data format and data specification documents.
    e. Description of instrument(s) used in the data acquisition and their sampling characteristics.
    f. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

4. All GNSS-R observations shall be from GPS and from other systems (i.e., GLONASS, BeiDou, and Galileo) or as specified in each data Task Order RFP. GNSS-R observations collected from other Signal of Opportunity (SoOP) sources may be included as additional data as specified in a data Task Order RFP.

5. Navigation solutions calculated onboard the spacecraft shall be provided, including receiver clock offsets and time-tagged Earth-Centered-Earth-Fixed position and velocity.

6. Navigation solution data shall be provided at intervals of 30 seconds or less.

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7. Satellite attitude quaternions shall be collected concurrently with all data at intervals of 30 seconds or less.

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FC3: Microwave Sounder Requirements

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C.6. Functional Category 3: MICROWAVE SOUNDER DATA IDIQ REQUIREMENTS

C.6.1. INTRODUCTION

The National Oceanic and Atmospheric Administration (NOAA) depends on rapid, high-quality environmental observations to improve weather forecasting and support mission-critical decision-making. Microwave (MW) sounding data provide all-weather measurements of atmospheric temperature and moisture structure as well as land and hydrology, which are essential inputs to numerical weather prediction (NWP) and short-term hazard monitoring.

Commercial MW sounder systems offer an opportunity to enhance these capabilities by delivering high-revisit, low-latency observations that can fill temporal gaps in the current observing system. In addition to traditional MW sounding channels, emerging commercial instruments may incorporate new techniques and novel spectral channels that improve sensitivity to boundary-layer processes, strengthen vertical retrievals, and provide additional information content for data assimilation. These innovations can directly improve forecast accuracy and support rapid-update applications where frequent atmospheric sampling is critical.

C.6.2. SCOPE

Under this acquisition, NOAA seeks access to commercial, near-real-time microwave radiance data from on-orbit Microwave Sounder (MWS) systems, including hyperspectral MWS observations, to support NOAA weather forecasting and broader environmental monitoring. The scope focuses on acquiring high-revisit, all-weather microwave sounding measurements that provide atmospheric temperature and moisture information at local, regional, and global scales. Emphasis is placed on data suitable for operational data assimilation, short-term forecast improvement, tropical cyclone tracking and intensity estimation, precipitation-related applications, and environmental hazard monitoring. Depending on the sensor design, MW sounders may also contribute to foundational MW imagery, and atmospheric vertical wind profiles.

This acquisition provides NOAA with scalable and flexible access to commercial microwave radiance data, enhancing forecast accuracy, improving situational awareness, and supporting NOAA's role as the Nation's authoritative source of environmental intelligence.

C.6.3. DATA REQUIREMENTS

All delivered data, including metadata and ancillary files, must meet NOAA standards or be fully compatible with NOAA operational systems, formats, and protocols for seamless integration, processing, and archival.

C.6.3.1. DEFINITIONS

The following definitions are provided in the context of all Microwave Sounder measurements:

1. Metadata are structured information that describe a dataset's key characteristics, including its content, origin, format, spatial and temporal coverage, processing history, and usage constraints. They provide essential context for data discovery, interpretation, integration, and reuse. High-quality metadata ensures transparency, traceability, and interoperability across NOAA's data enterprise and the broader scientific community.
   For additional guidance on metadata standards and practices, see NOAA's National Centers for Environmental Information (NCEI): https://www.ncei.noaa.gov/resources/metadata/introduction-to-metadata-at-ncei

2. NOAA Data Processing Level Definitions:

Table 1. NOAA Data Processing Level

Data Level	Description	Microwave Sounder Examples
Level 0	Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed.	Raw detector counts from each microwave channel, time stamps for each scan line, scan geometry and sensor view angles, instrument telemetry.
Level 1A (L1A)	Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information. This ancillary information can include instrument status, radiometric and geometric calibration coefficients and georeferencing parameters (e.g., platform ephemeris).	Uncalibrated digital counts from ATMS or AMSU channels with time, scan angle, and instrument telemetry.
Level 1B (L1B)	Calibrated and geolocated radiance data at full resolution in physical units such as brightness temperatures. Sensor measurements are converted using onboard calibration parameters.	Brightness temperature measurements from ATMS or AMSU at each channel, mapped to Earth location (latitude, longitude, scan time).

Level 1C (L1C)	Resampled and geolocated brightness temperature data projected onto a standard Earth grid, enabling multi-sensor analysis and model ingestion.	ATMS brightness temperatures gridded to a fixed Earth coordinate system (e.g., EASE Grid) for direct model assimilation.
Level 2 (L2)	Geophysical variables derived from L1B or L1C data at the sensor's native resolution. These are science-quality products used in weather and climate applications.	Vertical temperature and moisture profiles, surface temperature, or precipitation rate.

AMSU: Advanced Microwave Sounding Unit
ATMS: Advanced Technology Microwave Sounder
EASE: Equal Area Scale Earth

3. The latency of a data product is defined as the difference between the time the instrument acquires data in orbit and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).

4. Satellite data availability is the percentage of satellite-generated data that, during a given time period, is accessible, usable, and delivered as expected to users. It reflects the overall capability of the satellite system—including onboard sensors, communication links, ground stations, and processing infrastructure—to provide data continuously and without interruption.

C.6.3.2. OBSERVATION and DATA CAPABILITY AND REQUIREMENTS

1. The Contractor shall supply MWS observation data and derived products collected from on-orbit satellites. The observation data will primarily be used for atmospheric vertical temperature and moisture profiles measurements as well as for tropical cyclone observations.

2. The requirements set forth below constitute the specific operational measurement requirements. Requirements are categorized as threshold values, representing minimum acceptable levels, and objective values, representing desired levels of performance with increased utility. Individual Task Order solicitations may include updated requirements.

3. Geophysical products that may be derived from MWS data include, but are not limited to:
    a. Atmospheric Vertical Temperature Profile (AVTP) - All Weather
    b. Atmospheric Vertical Moisture Profile (AVMP) - All Weather
    c. Precipitation - contributor
    d. Atmospheric Vertical Wind Profile - contributor

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e. Select Land & Hydrology Products - contributor

4. Table 2 serves as general guidance for the desired spectral coverage for Earth microwave sounder data. These regions are based on prior Government analyses and are intended to support the use of advanced microwave sensor technologies while mitigating risks associated with microwave radio-frequency interference (RFI).

   Contractors shall select channels to maximize information content, with bandwidths appropriate for achieving the required range and vertical resolution of moisture and temperature sounding. For temperature- and moisture-sounding channels, bandwidths should generally align with those used by ATMS for comparable channels. Additional channels may be used to increase vertical resolution through appropriately spaced weighting functions.

   All spectral acquisitions shall remain within Earth exploration-satellite service (EESS) allocations, as defined in the FCC Online Table of Frequency Allocations, to ensure regulatory compliance and maintain system performance. The Government may identify additional or more specific spectral requirements in a Task Order RFP.

Table 2. Spectral Guidance for MW Radiance Observations

Spectral Channels	**Observation**	**Primary Use**
Ka, K (Window)	Cloud, Ice, Precipitation	Measure surface emissivity, Total Precipitable Water (TPW) and cloud liquid water
V, F (Temperature)	V band (50 – 60 GHz) and F band near 118 GHz contain O_2 absorptions sensitive to temperature profiling; 52-54 GHz crucial for determining precipitation type (rain vs. snowfall)	Vertical profiles of atmospheric temperature; temperature mapping as a function of altitude, includes channels to measure surface temperature
Channels for MW Window region	One or more channels near 23.8, 31.4, 88.2, 165.5, or 229 GHz	Channels for cloud, precipitation, and ice detection for direct all-sky assimilation and NWP Quality Control
G (Moisture)	G-band (~183 GHz) senses water vapor for profiling and ice scattering for precipitation	Vertical profiles of atmospheric moisture and relative humidity; moisture mapping at multiple altitudes; precipitation (rain and snowfall)
W, D (Window)	Earth's surface and/or precipitation observations	Sensing surface properties and scattering signals that are unique for acquiring precipitation, such as rain and snowfall rate

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J (Window)	Cloud, Ice, Precipitation	Experimental alternative to relatively lower frequencies with higher spatial resolution and better scattering contrast signals

5. Table 3 provides guidance for threshold and objective capabilities associated with hyperspectral MW Sounder capabilities with prioritization. These priorities are intended to help engineers and designers understand which sensor features have the greatest impact on global NWP performance. Each parameter includes three tiers — ideal (objective), nominal, and baseline (threshold) — to show the acceptable performance range. The tiers should be considered independently for each parameter unless otherwise noted; for example, selection of Tier 1 for one parameter does not require Tier 1 for others.

   This guidance is not meant to be restrictive. It is intended to support informed engineering decisions, allow flexibility for technology maturation, and enable cost-effective solutions while maintaining the performance necessary for operational weather prediction. Individual Task Orders will contain additional or more specific data acquisition requirements as needed.

Table 3. Guidance for Hyperspectral MW Sounder Capabilities
(prioritized from top to bottom)

Parameter	Tier 1 (Ideal)	Tier 2 (Nominal)	Tier 3 (baseline)
Temperature sounding channels	≥ 20 channels around 50–60 GHz and 118 GHz O_2 line	≥ 13 channels around 50–60 GHz	≥ 12 channels around 118 GHz O_2 line (TROPICS-type)
Moisture sounding channels	≥ 10 channels around 183 GHz	≥ 5 channels around 183 GHz	TEMPEST-type channels around 183 GHz or 325 GHz
Cloud/precip/ice detection channels	≥ 4 channels at 22–23 GHz, 31–37 GHz, 89 GHz, and 18 GHz if feasible	≥ 2 channels at 22–23, 31–37, 89, or 18–19 GHz	Channels in 200–300 GHz range
NEDT – Temperature sounding	ATMS-type or better	ATMS × 1.5	ATMS × 2
NEDT – Moisture sounding	ATMS-type or better	ATMS × 1.5	ATMS × 2
Spatial resolution	~ 15 km at nadir	~ 25 km at nadir	~ 50 km at nadir
Calibration	0.75–1 K (≤ 0.4 K	1–1.5 K (≤ 1 K non-linearity)	1–2 K (≤ 1.5 K

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accuracy	non-linearity)		non-linearity)
Temporal refresh	Every 2 hours	Every 3 hours	Every 6 hours

NEDT: Noise Equivalent Differential Temperature

TEMPEST: Temporal Experiment for Storms and Tropical Systems Technology

TROPICS: Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats

Hyperspectral capabilities are expected to provide enhanced radio frequency interference (RFI) detection. The frequency ranges listed in Table 4 include both the science function and the extended region of interest for RFI detection. These capabilities are provided as guidance only to inform industry of potential directions for future acquisitions and technology maturation.

Table 4: Prioritization of MW Sounder Hyperspectral Capabilities

Frequency Bands[1]	**Hyperspectral Capability**	**RFI Detection Capability**	**Hyperspectral Resolution (MHz)**
K (20-25 GHz)	Objective	Baseline[4]	≤ 10[5]
Ka (31-33 GHz)	Objective	Baseline[4]	≤ 10[5]
V (50-62 GHz)	Baseline[2]	Baseline[4]	≤1 between 56-58 GHz ≤ 10 outside of 56-58 GHz[5]
W (85-92 GHz)	Baseline	Baseline[4]	≤ 20[5]
F (108-123 GHz)	Baseline[2]	Objective	≤ 20
D (164-167 GHz)	Objective	Objective	N/A
G (173-193 GHz)	Baseline[3]	Objective	≤ 100
J (200-235 GHz)	Objective	Objective	N/A

Note 1: Instrument design may cover a subset of frequency range associated with a letter designation. The frequency range is not prescribing the precise start and stop frequencies or bandwidths.

Note 2: Capability required for Temperature Sounding (Table 3).

Note 3: Capability required for Moisture Sounding (Table 3).

Note 4: RFI detection capability required for continuous operational use.

Note 5: RFI detection capability required for diagnostic purposes only.

6. Table 5 summarizes the observational requirements for microwave (MW) sensors that most directly affect temperature and moisture sounding performance. The NEDT guidance here refers to the overall noise level of the channels. Particular attention

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should also be paid to the striping effect, which should be minimized as much as possible (or eliminated) when designing the sensor.

The Contractor shall meet or exceed these requirements, with the Government reserving the option to specify more detailed thresholds in a Task Order RFP.

When global coverage is not required, the Government may define regional coverage in a Task Order. The Contractor shall provide data covering at least 75% of the specified geographical region, averaged over the platform's orbital coverage for the designated period. Measurement stability and the ability to maintain required performance over the sensor's operational lifetime are essential.

Table 5. MW Sounder Observational and Data Requirements

Requirement Type	Performance Area	Threshold Requirement	Reference
Observation	Channels for direct Temperature Sounding	Spectral channels enabling direct temperature sounding	Table 2
	Channels for direct Moisture Sounding	Spectral channels enabling direct moisture sounding	Table 2
	Channels for cloud, precipitation, ice detection for direct all-sky assimilation and NWP	Spectral channels supporting cloud, precipitation, and ice detection for direct all-sky assimilation and NWP	Table 2
Data	Nadir Horizontal Resolution	≤ 32 km for Temperature ≤ 16 km for Moisture	—
	L1B product Latency	≤ 60 minutes	—
	Refresh Rate	≤ 6 hours	—
	NEDT for Temperature sounding channels	Better than ATMS-level NEDT x 2	—
	Noise level NEDT for Moisture sounding channels	Better than ATMS-level NEDT x 2	—
	Radiometric Calibration accuracy	1 K - 2 K depending on channels	—

C.6.4. DELIVERY REQUIREMENTS

C.6.4.1. DELIVERY REQUIREMENTS

Contractors shall supply data and metadata that meets the following requirements. The Government may stipulate more specific delivery requirements in a Task Order RFP.

1. Unless otherwise specified in a Task Order, the Contractor shall deliver Level 1 data products in formats and with data content comparable to NOAA's established data standards and specifications, such as those used for ATMS, to ensure compatibility with NOAA's operational processing and assimilation systems.

2. The Contractor shall maintain at least 75% operational availability, unless otherwise specified in a Task Order. This availability accounts for routine maintenance, calibration activities, and orbital dynamics, and ensures that the spacecraft are capable of supporting NOAA's data acquisition needs.

3. The Contractor shall, to the greatest extent possible, maintain continuous and contiguous near-real-time delivery of data collected during science observations to ensure a complete and uninterrupted data record. At a minimum, the Contractor must have the capability to provide continuous and contiguous data along-track for one satellite. One satellite of data represents the minimum ordering quantity for Task Orders.

4. Unless otherwise specified in a Task Order, the product latency of microwave soundings delivered by the Contractor, when averaged over any continuous thirty-day period, shall meet or exceed the required latency as specified in Table 5.

5. The Contractor shall ensure cross-platform data consistency across their constellation of commercial microwave sounder instruments. All instruments shall support "mix-and-match" interoperability, enabling seamless integration of radiance data regardless of satellite platform.

6. In the event of data anomalies or as requested by the Government, the Contractor shall provide Level 0 data spanning 1 to 3 months to support diagnostic analysis and root cause investigation, as well as independent calibration/validation by the Government or its designated representatives.

7. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. At a minimum, this shall include:

    a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.

    b. Documentation and data files to support characterization, quality control, and calibration/validation activities, including but not limited to:

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    i. Spectral response functions.
    ii. Polarization information.
    iii. Limb correction coefficients.
    iv. Community Radiative Transfer Model (CRTM) coefficients for each sensor.

c. Pre-launch and on-orbit calibration reports.
d. Documentation of quality control flags and calibration/validation indicators.
e. User guides and detailed data specification documents.
f. Description of instrument(s) used in the data acquisition and their sampling characteristics.
g. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

C.6.4.2. OPTIONAL DELIVERY REQUIREMENTS

C.6.4.2.1. BUFR Data

1. The contractor shall provide data in Binary Universal Form for the Representation of meteorological data (BUFR) format for L1B data. The BUFR data shall conform to current WMO standards and meet all applicable NOAA operational data formatting and ingest requirements.

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FC4: Multi-Spectral Imagery Requirements

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C.7. Functional Category 4: MULTISPECTRAL VISIBLE/INFRARED IMAGERY DATA IDIQ REQUIREMENTS

C.7.1. INTRODUCTION

Multispectral visible and infrared (VIS/IR) imagery, including low-light observations from the Day-Night Band (DNB), is a core observational capability supporting NOAA's mission to monitor and predict weather, climate, and environmental conditions. These data are fundamental to operational forecasting, hazard detection, and situational awareness, helping protect life, property, and economic activity.

Mission-critical VIS/IR imagery includes weather, cloud, water vapor, low-light, and wildfire imagery observations that enable continuous monitoring of atmospheric and surface conditions across day, night, and high-latitude regions. Low-light imagery is especially valuable in polar areas, including Alaska, which are not fully covered by geostationary satellites.

To ensure continuity, resilience, and flexibility, NOAA is strategically incorporating commercial multispectral VIS/IR imagery, including low-light data, to complement government satellite systems. Commercial data can expand coverage, improve revisit times, accelerate adoption of new technologies, and reduce operational risk during satellite transitions or anomalies, supporting a diversified and resilient Earth observation architecture.

C.7.2. SCOPE

Under this acquisition, NOAA seeks access to commercial multispectral visible, infrared, and low-light satellite imagery from on-orbit systems to support NOAA weather forecasting and broader environmental monitoring. The scope focuses on weather, cloud, low-light, and fire imagery acquisition at local, regional, and global scales, with emphasis on cloud and fog detection, wildfire detection and characterization, basic scene classification, and nighttime and high-latitude coverage. Value-added or derived imagery products may be included as needed to meet mission objectives, as defined in individual Task Orders.

This acquisition provides NOAA with scalable and flexible access to commercial VIS/IR and low-light imagery, strengthening observational continuity, enhancing situational awareness, and supporting NOAA's role as the Nation's authoritative source of environmental intelligence.

C.7.3. DATA STANDARD AND COMPATIBLE REQUIREMENTS

All delivered data, including metadata and ancillary files, must meet NOAA standards or be fully compatible with NOAA operational systems, formats, and protocols for seamless integration, processing, and archival.

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C.7.3.1. DEFINITIONS

The following definitions are provided in the context of all Multispectral Imagery measurements:

1. Metadata are structured, descriptive information that document the characteristics of a dataset. They provide essential context about the data's content, origin, purpose, quality, format, spatial and temporal coverage, processing history, and usage constraints. Metadata enables effective data discovery, interpretation, integration, management, and reuse by both humans and machines. High-quality metadata are critical for ensuring transparency, traceability, and interoperability within NOAA's data enterprise and across the broader scientific community. For additional guidance on metadata standards and practices, see NOAA's National Centers for Environmental Information (NCEI): https://www.ncei.noaa.gov/resources/metadata/introduction-to-metadata-at-ncei

2. NOAA Data Processing Level Definitions:

Table 1. NOAA Data Processing Level

Data Level	Description	Multispectral Imagery Examples
Level 0 (L0)	Level-0 (L0) data consist of unprocessed, full-resolution instrument measurements as received from the satellite, including raw detector counts and ancillary telemetry needed for processing.	Raw detector counts from each channel, time stamps for each scan line, scan geometry and sensor view angles, instrument telemetry.
Level 1A (L1A)	Level-1A (L1A) data are reconstructed and time-ordered instrument measurements that have been de-packetized and quality-checked but remain uncalibrated.	Uncalibrated digital counts from each channel with time, scan angle, and instrument telemetry.
Level 1B (L1B)	Level-1B (L1B) data contain radiometrically calibrated and geolocated radiance measurements for each spectral band. Processing includes: conversion from raw counts to physical radiance units; application of calibration coefficients; assignment of Earth-location coordinates (geolocation) and quality flags and metadata. L1B data are the primary input for generating geophysical products.	Calibrated, geolocated radiances or reflectances for each spectral band and spatial pixel.
Level 2 (L2)	Level-2 (L2) data are derived geophysical products — such as cloud properties, sea-surface temperature, aerosol optical depth, fire detections, and vegetation	Cloud properties, sea-surface temperature, aerosol optical depth, fire detections

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	indices — computed from Level-1B radiances and ready for scientific and operational use.	

3. The latency of a data product is defined as the difference between the time the instrument acquires data in orbit and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).

4. Satellite data availability is the percentage of satellite-generated data that, during a given time period, is accessible, usable, and delivered as expected to users. It reflects the overall capability of the satellite system—including onboard sensors, communication links, ground stations, and processing infrastructure—to provide data continuously and without interruption.

C.7.3.2. OBSERVATION CAPABILITY AND REQUIREMENTS

C.7.3.2.1. Observation Capability and Requirements - All Areas

1. The Contractor shall provide multispectral observation data collected from on-orbit satellites. Derived products may also be required when specified in individual Task Order solicitations.

2. The requirements described below define the baseline operational measurement expectations. These requirements may be refined or updated in individual Task Order solicitations as mission needs evolve.

3. All spectral acquisitions shall comply with Earth Exploration-Satellite Service (EESS) frequency allocations, as defined in the FCC Online Table of Frequency Allocations, to ensure regulatory compliance and maintain system performance integrity.

4. The spectral coverage reference, summarized in Table 2, reflects capabilities demonstrated across multiple legacy missions. Ideally, systems shall provide at least three bands spanning the visible, near-infrared (VNIR, 0.4 – 1 µm) and shortwave infrared (SWIR, 1 – 3 µm) regions; a minimum of two midwave infrared (MWIR, 3–8 µm) bands, including at least one water-vapor-sensitive channel; and provide longwave infrared (LWIR, 8 - 14 µm) coverage centered near 10.8 µm. However, a subset of these bands or other overlapping channels systems could support weather-focused imagery to meet mission needs.

   Table 2 is intended to ensure continuity with heritage measurements and is therefore guidance. In disaggregated sensor architectures, Contractors may focus individual instruments on specific product families and perform trade studies to determine the optimal channel set needed to achieve required accuracy and precision for those

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products. As Table 2 provides overarching guidance, Contractors would need to meet, at a minimum, one or more of the spectral ranges as identified in Table 2 for a capability area. The Government will specify more specific guidance in a Task Order RFP.

Table 2. Spectral reference for Weather, Cloud, Fire and Water Vapor Imaging capabilities

Legacy Sensors (for reference)	**Objective Spectral Range (µm)**	**Capability Areas:**			
		Weather Imagery	**Cloud**	**Fire**	**Water Vapor**
VIIRS, OLS (DNB)	0.5 - 0.9	x	x		
VIIRS, MODIS	0.40 - 0.42		x		
VIIRS, AVHRR, MODIS, OLS	0.60 – 0.68	x		x	
VIIRS, MODIS	0.66 - 0.68		x	x	
VIIRS, MODIS	0.73 - 0.75				
VIIRS, AVHRR	0.84 - 0.89	x		x	
MODIS, METImage	0.89 – 0.92				x
VIIRS, MODIS	1.23 - 1.25		x		
VIIRS, MODIS	1.37 - 1.39		x		
VIIRS, AVHRR, MODIS	1.58 - 1.64	x		x	
VIIRS	2.23 - 2.28		x	x	
VIIRS, AVHRR, MODIS	3.55 – 3.93	x		x	
VIIRS, MODIS	3.66 - 3.84		x		
VIIRS, MODIS	3.97 - 4.13		x	x	
GXI	5.1-5.2				x
ABI	5.7-6.6				x
MODIS	6.5-6.9				x
METImage	6.7-7.2				x
MODIS, METImage	7.1-7.5				x
VIIRS, MODIS	8.4 - 8.7	x	x		
VIIRS, AVHRR, MODIS	10.26 - 11.26	x	x	x	
VIIRS, OLS	10.5 – 12.4	x		x	
VIIRS, AVHRR, MODIS	11.54 - 12.49	x	x	x	

ABI: Advanced Baseline Imager

AVHRR: Advanced Very High Resolution Radiometer

GXI: GeoXO Imager

METImage: Meteorological Imager

MODIS: Moderate-resolution Imaging Spectro-radiometer

OLS: Operational Linescan System

VIIRS: Visible Infrared Imager Radiometer Suite

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C.7.3.2.2. Observation Capability and Requirements - Weather, Cloud and Low-Light Imagery

1. Low-light weather imaging over Alaska is a mission-critical driver for constellation design. NESDIS considers DNB capability essential due to its proven value in this region. The Government will specify more specific guidance in a Task Order RFP.
2. To support NOAA's geophysical observation capabilities, the general performance performance requirements for a multispectral imager are listed in Table 3. The Government may specify more stringent requirements in a Task Order RFP.

Table 3. Fundamental Performance Requirements for Multispectral Imagery

Performance Category	Requirement
Spatial Resolution	Instantaneous Field of View (IFOV) Resolution shall not differ by more than 2x between the edge of scan and nadir
Spectral Response	Ensure sensor relative spectral responses are well-characterized and calibrated
Radiometric Calibration	Ensure consistency in radiometric accuracy, dynamic range, SNR (signal-to-noise ratio), and/or Noise Equivalent Differential Temperature (NEDT) with legacy sensors for the same application targets

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3. Table 4, summarized below, defines the observation requirements for weather and cloud imagery, including low-light DNB imagery. The Contractor shall provide weather imagery across the visible, near-infrared (NIR), IR, and low-light spectral regions that meet one or more of the specified spectral ranges. The Government may further refine these requirements, including specifying targeted spectral ranges, observation quantities, swath, resolution, sensitivity, accuracy, latency, refresh and coverage, within individual Task Order solicitations.

   When global coverage is not required, the Government may define regional coverage in a Task Order. The Contractor shall provide data covering at least 75% of the specified geographical region, averaged over the platform's orbital coverage for the designated period. Measurement stability and the ability to maintain required performance over the sensor's operational lifetime are essential.

Table 4. Multispectral Observation Requirements for Weather, Cloud and Low-Light Imagery

Performance Level	Threshold Requirement
Horizontal Spatial Resolution	750 m for M-Band, 375 for I-Band at nadir,
Low Light Spatial Resolution	750 m at nadir
Geolocation Accuracy	50% of Horizontal spatial resolution
Low Light Sensitivity	50% Lunar illumination
L1B product Latency	60 minutes (Global) 30 minutes (Polar-Alaska)
Coverage	Regional, global, or polar area (60°-90° N, 170°-135° W)

Note: 1. Polar-Alaska Region is defined as the area bounded by 60°N to 90°N and 170°W to 135°W

C.7.3.2.2.1. Future Capability Considerations for Weather, Cloud and Low-light Imagery

1. In addition to the baseline requirements and references outlined in Tables 2–4, several aspirational improvements have been identified to enhance future imaging capabilities. For the Day–Night Band (DNB), these include improved spatial resolution to better support low-light observations, particularly over high-latitude regions such as Alaska, and expanded spectral coverage into the blue region (~0.4 µm) to enable improved detection of light-emitting diode (LED) lighting. For cloud and weather imagery products, aspirational enhancements include the addition of water vapor channels, with bands centered near 0.91 µm, 5.15 µm, 6.2 µm, 6.9 µm, and 7.3 µm to better distinguish moisture features at varying atmospheric altitudes. Expanded spectral resolution is also desired to improve discrimination among cloud types and other scene characteristics. Since these are aspirational, they do not represent a requirement for Contractors, but provide guidance for future improvements.

C.7.3.2.3. Observation Capability and Requirements - Fire Imagery

1. Spectral coverage guidance for fire imagery is informed by heritage mission performance and summarized in Table 2. Table 5, summarized below, presents the observational requirements for fire imagery detection. The Contractor shall comply with these requirements and meet or exceed the stated thresholds. While the objective requirements in the Table 5 reflect current technological capabilities, data at 10–50 m resolution is particularly valuable when active fire targets are smaller than the native

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pixel scale. The Government may provide more detailed or refined thresholds in individual Task Order solicitations.

When global coverage is not required, the Government may define regional coverage in a Task Order. The Contractor shall provide data covering at least 75% of the specified geographical region, averaged over the platform's orbital coverage for the designated period. Measurement stability and the ability to maintain required performance over the sensor's operational lifetime are essential.

Table 5. VNIR High Refresh Rate Land Observational Requirements

Performance Level	Threshold Requirement
Horizontal Spatial Resolution	750 m threshold, 200 m objective
Geolocation Accuracy	50% of Horizontal spatial resolution
L1B data Latency	2 hr
Coverage	Regional, global, or polar area (60°-90° N, 170°-135° W)

Note: 1. Coverage rate is evaluated at constellation level, not individual orbit plane level.

C.7.4. DELIVERY REQUIREMENTS

The Contractor shall provide data and metadata that meets the requirements outlined below. The Government may specify additional delivery requirements in individual Task Order solicitations.

1. Unless otherwise specified in a Task Order, the Contractor shall deliver Level 1 data products in formats and with data content comparable to NOAA's established data standards and specifications, such as those used for Visible Infrared Imaging Radiometer Suite (VIIRS), to ensure compatibility with NOAA's operational processing and assimilation systems.

2. The Contractor shall maintain at least 75% operational availability, unless otherwise specified in a Task Order. This availability accounts for routine maintenance, calibration activities, and orbital dynamics, and ensures that the spacecraft are capable of supporting NOAA's data acquisition needs.

3. The Contractor shall, to the greatest extent possible, maintain continuous and contiguous near-real-time delivery of data collected during science observations to ensure a complete and uninterrupted data record. At a minimum, the Contractor must

have the capability to provide continuous and contiguous data along-track for one satellite. One satellite of data represents the minimum ordering quantity for Task Orders.

4. Unless otherwise specified in a Task Order, the data latency delivered by the Contractor, when averaged over any continuous thirty-day period, shall meet or exceed the required latency as specified in Table 4 (for Weather, Clouds and Low-Light Imagery) and Table 5 (for Fire Imagery).
5. The Contractor shall ensure cross-platform data consistency across their constellation of commercial infrared sounder instruments. All instruments shall support "mix-and-match" interoperability, enabling seamless integration of radiance data regardless of satellite platform.
6. In the event of data anomalies or as requested by the Government, the Contractor shall provide Level 1A data spanning 1 to 3 months to support diagnostic analysis and root cause investigation and/or to perform independent calibration/validation by the Government or its designated representatives.
7. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. At a minimum, this shall include:
    a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.
    b. Documentation and data files to support instrument characterization, quality control, and calibration/validation activities, including but not limited to:
        i. Spectral response functions.
        ii. Polarization information.
        iii. Gain and offset values for each spectral band.
        iv. Nonlinearity information.
        v. Temperature-dependent calibration curves.
        vi. Dark current and bias corrections.
        vii. Cross-talk or stray-light information.
        viii. System/optical degradation information
        ix. Band-to-Band registration information
    c. Pre-launch and on-orbit calibration reports.
    d. Documentation of quality control flags and calibration/validation indicators.
    e. Data specification documents and user guides, including file name conventions.

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f. Description of instrument(s) used in the data acquisition and their sampling characteristics.

g. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

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FC5: Infrared Sounder Requirements

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C.8. Functional Category 5: INFRARED SOUNDER DATA IDIQ REQUIREMENTS

C.8.1. INTRODUCTION

NOAA seeks to acquire commercial, near-real-time hyperspectral infrared (IR) sounder data from on-orbit systems to enhance operational weather forecasting, numerical weather prediction (NWP), and environmental monitoring. Hyperspectral IR sounders provide high-vertical-resolution information on atmospheric temperature, moisture, and cloud properties, which is essential for improving short-term forecasts, severe weather readiness, and global environmental intelligence. Access to commercial IR sounder data increases observational resilience, expands revisit opportunities, and strengthens NOAA's ability to deliver timely and accurate environmental information.

Infrared (IR) Sounder observations are essential to NOAA's mission. They provide critical inputs for numerical weather prediction (NWP), hydrological modeling, weather monitoring, tropical cyclone intensity estimation, and disaster response efforts. IR sounders enable retrievals to atmospheric vertical temperature and moisture profiles and atmospheric chemistry specifications. IR soundings support applications for land surface and cloud characterization, atmospheric composition, and radiation budget.

C.8.2. SCOPE

Under this acquisition, NOAA seeks access to commercial hyperspectral IR sounding data delivered in near real time to support operational data assimilation, forecast modeling, and environmental analysis. The scope focuses on obtaining IR sounding data across the longwave, midwave, and shortwave infrared regions to provide detailed atmospheric profile information at local, regional, and global scales.

NOAA seeks IR sounder data to expand observational capacity, increase temporal refresh, and broaden spatial coverage available to operational systems. Value-added or derived products may be included when needed to meet mission objectives, as defined in individual Task Orders. This acquisition provides NOAA with scalable and flexible access to commercial IR sounder data, strengthening observational continuity and supporting NOAA's role as the Nation's authoritative source of environmental intelligence.

C.8.3. DATA REQUIREMENTS

All delivered data, including associated metadata and ancillary files, shall conform to NOAA data standards or be fully compatible with NOAA operational data systems, formats, and protocols, to ensure seamless integration, processing, and archival within NOAA's enterprise systems.

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C.8.3.1. DEFINITIONS

The following definitions are provided in the context of all Infrared Sounder measurements:

5. Metadata are structured, descriptive information that document the characteristics of a dataset. They provide essential context about the data's content, origin, purpose, quality, format, spatial and temporal coverage, processing history, and usage constraints. Metadata enables effective data discovery, interpretation, integration, management, and reuse by both humans and machines. High-quality metadata are critical for ensuring transparency, traceability, and interoperability within NOAA's data enterprise and across the broader scientific community.
   For additional guidance on metadata standards and practices, see NOAA's National Centers for Environmental Information (NCEI): https://www.ncei.noaa.gov/resources/metadata/introduction-to-metadata-at-ncei

6. NOAA Data Processing Level Definitions:

Table 1. NOAA Data Processing Level

Data Level	Description	Infrared Sounder Examples
Level 0	Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed.	Raw detector counts from each Infrared channel, time stamps for each scan line, scan geometry and sensor view angles, instrument telemetry.
Level 1A (L1A)	Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information. This ancillary information can include instrument status, radiometric and geometric calibration coefficients and georeferencing parameters (e.g., platform ephemeris).	Uncalibrated digital counts from CrIS channels with time, scan angle, and instrument telemetry.
Level 1B (L1B)	Calibrated and geolocated radiance data at full resolution in physical units such as brightness temperatures. Sensor measurements are converted using onboard calibration parameters.	Brightness temperature measurements from CrIS at each channel, mapped to Earth location (latitude, longitude, scan time).
Level 1C (L1C)	Resampled and geolocated brightness temperature data projected onto a standard Earth grid, enabling multi-sensor analysis and model ingestion.	CriS brightness temperatures gridded to a fixed Earth coordinate system (e.g., EASE-Grid) for direct model assimilation.

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Data Level	Description	Infrared Sounder Examples
Level 0	Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed.	Raw detector counts from each Infrared channel, time stamps for each scan line, scan geometry and sensor view angles, instrument telemetry.
Level 1A (L1A)	Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information. This ancillary information can include instrument status, radiometric and geometric calibration coefficients and georeferencing parameters (e.g., platform ephemeris).	Uncalibrated digital counts from CrIS channels with time, scan angle, and instrument telemetry.
Level 2 (L2)	Geophysical variables derived from L1B or L1C data at the sensor's native resolution. These are science-quality products used in weather and climate applications.	Vertical temperature and moisture profiles, outgoing longwave radiation, or atmospheric composition measurements derived from CrIS radiances.

CrIS: Cross-track Infrared Sounder
EASE: Equal Area Scale Earth

7. The latency of a data product is defined as the difference between the time the instrument acquires data in orbit and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).

8. Data availability refers to the consistent, uninterrupted delivery of data to NOAA and the maintenance needed to ensure required data delivery specifications.

C.8.3.2. OBSERVATION CAPABILITY AND REQUIREMENTS

7. The Contractor shall supply IR Sounder observation data and derived products collected from on-orbit satellites. The observation data will primarily be used for atmospheric vertical temperature and moisture profiles measurements as well as for other cloud characterization property and atmospheric composition measurements.

8. The requirements set forth below constitute the specific operational measurement requirements. Requirements are categorized as threshold values, representing minimum acceptable levels, and objective values, representing desired levels of performance with increased utility. Individual Task Order solicitations may include updated requirements.

9. Geophysical products that may be derived from IR Sounder data include, but are not limited to:
    a. Atmospheric Vertical Temperature Profile (AVTP) - Cloud-free to partly cloudy

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b. Atmospheric Vertical Moisture Profile (AVMP) - Cloud-free to partly cloudy
c. Atmospheric Composition Products
d. Radiation Budget - Outgoing Longwave Radiation
e. Atmospheric Vertical Wind Profile (to include 3D Winds) - contributor

10. Table 2, summarized below, lists the general spectral guidance and primary detection utility for IR Radiance observations. While Table 2 provides overarching guidance, Contractors shall provide, at a minimum, IR Radiance observations that satisfy capabilities within one or more of the spectral ranges as identified in Table 2. Spectral acquisitions must remain within EESS (Earth exploration-satellite service) allocations, as specified in the FCC Online Table of Frequency Allocations, to avoid compromising system performance or violating regulatory constraints. The Government may stipulate more specific guidance, to include more targeted spectral ranges, in a Task Order RFP.

Table 2. Spectral Guidance for IR Radiance Observations

Spectral Band	**Spectral Channel Ranges (μm)**	**Primary Detection Use**
Shortwave Infrared	1.5 - 4.8	Surface/cloud properties, atmospheric temperature, aerosol composition
Midwave Infrared	5.7 - 8.6	Atmospheric water vapor and temperature, aerosol composition
Longwave Infrared	8.5 - 15.4	Atmospheric temperature, cloudy sky retrieval on sensitivity to cold objects, cloud top height

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11. Table 3a, summarized below, presents the observational requirements for IR sensors that most strongly influence the performance of IR sounders in measuring atmospheric temperature and moisture and atmospheric composition. The Contractor shall adhere to these requirements and meet or exceed the stated thresholds. The Government may stipulate more specific threshold requirements in a Task Order RFP.

12. Table 3b, summarized below, lists general guidance for IR Sounder data. This guidance is not meant to be restrictive. It is intended to support informed engineering decisions, allow flexibility for technology maturation, and enable cost-effective solutions while maintaining the performance necessary for operational weather prediction. Individual Task Order RFPs will contain additional and/or more specific data acquisition requirements as needed.

    When global coverage is not required, the Government may define regional coverage in a Task Order. The Contractor shall provide data covering at least 75% of the specified geographical region, averaged over the platform's orbital coverage for the designated period. Measurement stability and the ability to maintain required performance over the sensor's operational lifetime are essential.

Table 3a. IR Sounder Data Requirements

Performance Factor	Threshold Requirement
Spatial Resolution (at Nadir)	15 km
Latency	60 minutes
Coverage	Global in 24 hours

Table 3b. IR Sounder Data Guidance

Performance Factor	Threshold
Spatial Resolution	0.625 cm^{-1} unapodized or better
Spectral Response Functions (or Instrument Line Shape)	Known with high accuracy (not allowed to limit Radiometric Calibration errors)
Spectral Calibration	5 parts per million (ppm) or better (Doppler correction must be included)
Spectral Sampling	Nyquist
Noise Equivalent Radiance	CrIS-type NEDT levels at CrIS spatial footprint (tolerance adjusted based on footprint)
Radiometric Calibration	Better than 0.2 K at 3-sigma brightness temperature
Spatial Coverage	Contiguous footprint, 10 km or better resolution
Geolocation and Cloud Detection Capability	1 km or better

NEDT: Noise Equivalent Differential Temperature

C.8.4. DELIVERY REQUIREMENTS

C.8.4.1. DELIVERY REQUIREMENTS

Contractors shall supply data and metadata that meets the following requirements. The Government may stipulate more specific delivery requirements in a Task Order RFP.

8. Unless otherwise specified in a Task Order, the Contractor shall deliver Level 1 data products in formats and with data content comparable to NOAA's established data standards and specifications, such as those used for CrIS, to ensure compatibility with NOAA's operational processing and assimilation systems.

9. The Contractor shall maintain at least 75% operational availability, unless otherwise specified in a Task Order. This availability accounts for routine maintenance, calibration activities, and orbital dynamics, and ensures that the spacecraft are capable of supporting NOAA's data acquisition needs.

10. The Contractor shall, to the greatest extent possible, maintain continuous and contiguous near‑real‑time delivery of data collected during science observations to ensure a complete and uninterrupted data record. At a minimum, the Contractor must have the capability to provide continuous and contiguous data along-track for one satellite. One satellite of data represents the minimum ordering quantity for Task Orders.

11. Unless otherwise specified in a Task Order, the product latency of infrared soundings delivered by the Contractor, when averaged over any continuous thirty-day period, shall meet or exceed the required latency as specified in Table 3a.

12. The Contractor shall ensure cross-platform data consistency across their constellation of commercial infrared sounder instruments. All instruments shall support "mix-and-match" interoperability, enabling seamless integration of radiance data regardless of satellite platform.

13. In the event of data anomalies or as requested by the Government, the Contractor shall provide Level 0 data spanning 1 to 3 months to support diagnostic analysis and root cause investigation and/or to perform independent calibration/validation by the Government or its designated representatives.

14. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. At a minimum, this shall include:

    a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.

    b. Documentation and data files to support characterization, quality control, and calibration/validation activities, including but not limited to:

        i. Spectral response functions.

        ii. Polarization information.

        iii. Limb effect correction coefficients.

        iv. Community Radiative Transfer Model (CRTM) coefficients for each sensor.

    c. Pre-launch and on-orbit calibration reports.

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d. Documentation of quality control flags and calibration/validation indicators.

e. User guides and detailed data specification documents.

f. Description of instrument(s) used in the data acquisition and their sampling characteristics.

g. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

C.8.4.2. OPTIONAL DELIVERY REQUIREMENTS

C.8.4.2.1. BUFR Data

2. The contractor shall provide data in Binary Universal Form for the Representation of meteorological data (BUFR) format for L1B data. The BUFR data shall conform to current WMO standards and meet all applicable NOAA operational data formatting and ingest requirements.

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FC6: Thermospheric Neutral Density Requirements

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C.9. Functional Category 6: THERMOSPHERIC NEUTRAL DENSITY DATA IDIQ REQUIREMENTS

C.9.1. INTRODUCTION

Thermospheric neutral density is a critical factor in the operation of satellites at Low Earth Orbit (LEO), as it directly affects the atmospheric drag experienced by satellites. These measurements play a crucial role for space situational awareness (SSA) operations such as collision avoidance, conjunction analysis, orbital decay, satellite control, and payload operations. As the satellite population at LEO dramatically increased over the previous solar cycle, there is a growing need to improve satellite environmental predictions to enhance NOAA's capability in space traffic coordination and ensuring safe satellite operations.

Neutral Density refers to the un-ionized particles of the upper atmosphere that can cause satellite drag. Thermospheric density is most significantly affected by solar activity and geomagnetic storms. Variations in thermospheric density affect drag forces acting on objects in LEO, impacting orbit propagation and prediction accuracy, satellite attitude control, and re-entry prediction. Accurate estimation of thermospheric neutral density is essential for precise orbit determination, collision avoidance, and re-entry predictions of satellites. Monitoring thermospheric density is critical to predict how quickly orbital decay occurs and to plan for deorbiting or orbital adjustments, and related data will help to improve neutral density nowcasts and forecasts that are critical for SSA operations in the LEO environment.

NOAA desires to acquire the data necessary to determine near real-time neutral density from the on-board tracking of active satellites. NOAA seeks to obtain Global Navigation Satellite System (GNSS) Precise Orbit Determination (POD) data, ephemeris data, attitude, and information about each satellite's physical characteristics. These detailed data are necessary for an accurate determination of the satellite drag environment.

C.9.2. SCOPE

NOAA is soliciting commercial, near-real-time satellite-based data to support operational data assimilation, forecast modeling, and environmental monitoring of the thermospheric density. NOAA may derive orbit-effective thermospheric neutral density internally using delivered orbit dynamics and spacecraft data, may acquire contractor-derived thermospheric density estimates as part of the delivered data, or may employ a combination of both approaches. NOAA will archive both the raw data and any derived datasets for operational use and long-term scientific research.

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C.9.3. DATA REQUIREMENTS

All delivered data, including associated metadata and ancillary files, shall conform to NOAA data standards or be fully compatible with NOAA operational data systems, formats, and protocols, to ensure seamless integration, processing, and archival within NOAA's enterprise systems. Data quantity, volume and delivery period duration will be specified in each Task Order RFP.

Each Task Order RFP may contain more specific data requirements to include, but not limited to, accuracy, coverage, data formats, latency, refresh, resolution, and sensitivity.

C.9.3.1. Definitions

The following definitions are provided in the context of Thermospheric Neutral Density measurements:

1. Latency: The time interval between an observation's time stamp and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).
2. Level 0 data: Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed. Level 0a (GNSS carrier phase, pseudorange), Level 0b (satellite ancillary data).
3. Level 1 data: Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information (such as PVT).
4. Level 2 data: Near real time (Level 2a) and derived POD data (Level 2b).
5. Level 3 data: Thermospheric density.
6. Non-Conservative Accelerations: Estimated accelerations not attributable to a central-body or third-body gravity. Depending on a vendor's OD configuration, these may include atmospheric drag, solar radiation pressure (SRP), Earth radiation pressure, thrust from maneuver or leakage, and/or any additional empirically estimated accelerations.
7. Orbit Arc: A continuous segment of orbit determination data spanning a fraction of an orbit to multiple orbits.
8. GNSS carrier phase (Level 0a): Raw GNSS data
9. Pseudorange (Level 0a): Raw GNSS data
10. Orbit Determination (OD) (Level 2): The process of estimating an orbiting object's position, velocity, and associated dynamic orbital parameters. Orbit Determination

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solutions may be produced using filtering, smoothing, or batch estimation techniques and necessarily include estimates of non-conservative accelerations such as atmospheric drag and solar radiation pressure.

11. Orbit-Effective Neutral Density (Level 3): A scalar estimate of the thermospheric neutral mass density derived from satellite orbit dynamics that represents the effective atmospheric density along the spacecraft trajectory that produces the observed change in orbital energy.

12. Position/Velocity/Time (PVT) (Level 1): Time-tagged satellite state vectors, including position and velocity, provided in an Earth-Centered Inertial (ECI) or Earth-Centered Earth-Fixed (ECEF) reference frame.

13. Satellite Operating State (Level 0b): Information describing spacecraft operational conditions, including maneuvers, thrusting, safe mode, payload operations, attitude transitions, or OD stabilization periods. Categorical representations are acceptable provided that they allow for the identification of periods of imposed orbital perturbations (e.g., thrust maneuvers, etc.) or non-ideal OD behavior.

C.9.3.2. Orbit Dynamics, Spacecraft, and Density Data

Contractors shall provide POD data sufficient to enable the determination of orbit-effective thermospheric neutral density. Such data shall consist of orbit dynamics and spacecraft information, contractor-derived density estimates, or a combination thereof, as specified in each Task Order RFP. The Government will use this data to process either POD antenna or raw GNSS information or both into orbit-effective thermospheric neutral density specifications. The Government may specify in each Task Order RFPs refined requirements.

For each participating satellite, Contractors shall provide:

1. Data and metadata suitable for POD (Level 0):
    - Dual frequency GNSS tracking data including pseudorange (Level 0a), carrier phase (Level 0a), and SNR observations, as collected on-orbit in real time, with a minimum data rate of 1 Hz.
    - Every orbit of each shall be covered by POD tracking data at no less than 75% duty cycle.
    - POD tracking data shall be from GPS satellites, plus optionally Galileo, GLONASS, and/or BeiDou satellites.
    - POD antenna reference point vector(s) relative to the center of mass and corresponding uncertainty estimates, antenna boresight vectors, and antenna fields of view (Level 0b).

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2. POD antenna phase center offset and phase center variation measurements and corresponding uncertainty estimates, in ANTEX format (https://files.igs.org/pub/data/format/antex14.txt) (Level 0b).
3. Navigation solutions calculated onboard the spacecraft for processing from GNSS data (Level 2a):
    - Navigation solutions shall include receiver clock offsets and time-tagged Earth-Centered-Earth-Fixed position and velocity.
    - Navigation solution data shall be provided at intervals of 60 seconds or less.
4. POD Outputs (Level 1):
    - Position and velocity ephemeris (PVT),
    - Time stamps in GPS seconds or UTC,
    - Reference frame definition and Earth orientation conventions.
5. OD-Estimated Non-Conservative Accelerations:
    - Atmospheric drag acceleration and equivalent drag-related or empirically estimated parameters,
    - Solar radiation pressure and other estimated or modeled accelerations and uncertainties, if available,
    - Clear identification of estimated versus modeled components.
6. Attitude, Configuration, and Operating State (Level 0b):
    - Spacecraft attitude (quaternions preferred, Euler angles allowed if exact order convention is specified) at intervals of 30 sec or less,
    - Panel articulation and configuration state,
    - Indicators for maneuvers, thrusts, OD filter stabilization periods, or other non-nominal operations (obfuscated or categorical representations are acceptable to protect proprietary operations),
    - Definition of the satellite body-fixed (SBF) coordinate system relative to the coordinate frame of the PVT/acceleration vectors and a detailed description of the application of attitude and panel articulation data relative to the SBF coordinate system.
7. Satellite Physical Characteristics (Level 0b):
    - Nominal wet and dry mass, as well as proxies for any anticipated change in wet mass,
    - Satellite geometry computer-aided design (CAD) model of all outer surfaces of the satellite.

While not required, for each participating satellite, Contractors may provide:

1. Derived Thermospheric Density (if provided) (Level 2):
    - Orbit-effective density data and associated time tags referencing the start and end times of the orbit,

- Associated uncertainty estimates, quality metrics, or flags identifying periods affected by maneuvers or non-nominal operations.

C.9.3.3. Temporal Resolution and Continuity

Contractors shall provide solutions necessary for temporal resolution and data continuity. The Government may stipulate further specifications in future Task Order RFPs.

For each participating satellite, Contractors shall provide:

1. Data at a nominal cadence of at least once per minute for processing from OD outputs.
2. Time stamps for PVT, non-conservative accelerations, and start/end times for any provided density estimates shall be uninterpolated and aligned with the native time step of the OD filter.
3. Attitude, panel articulation, and operating state data at each of their native time stamps (i.e., with no interpolation if possible).
4. Orbit Arcs:
    - Orbit arcs shall be for a duration of no less than 30 minutes with no gaps within an orbit arc exceeding 5 minutes. If gaps exist between adjacent orbit arcs, then attitude and configuration data should span these gaps.

C.9.3.4. Accuracy and Quality Characterization

Rather than prescribing a fixed accuracy threshold, Contractors shall provide self-characterized performance metrics describing the quality of delivered orbit, attitude, acceleration, and/or density data. The Government may stipulate further specifications in future Task Order RFPs.

At minimum, Contractors shall document:

1. Orbit Solution Quality
    - Estimated position and velocity uncertainty (e.g., RMSE, formal covariance, or equivalent),
    - OD methodology (e.g., batch, filter, smoother, etc.),
    - Conservative force model (e.g., gravity model and degree/order, third-body models and source),
    - Tracking data sources utilized.
2. Attitude Knowledge
    - Estimated attitude uncertainty,
    - Update rate and latency.
3. Fidelity of Derived Thermospheric Density (if provided)
    - Window length of estimation,
    - Detailed description of the physical model used to compute the drag and lift coefficients, along with input parameters, at each PVT time step.

C.9.3.5. Alternative Thermospheric Neutral Density Solutions

NOAA may consider alternative technological solutions to characterize thermospheric neutral density to include, but not limited to, accelerometers, mass spectrometers, and ion gauges. Contractors aspiring to utilize alternative technologies shall describe methodologies and demonstrate validation processes to enable the determination of thermospheric neutral densities. In addition, Contractors shall describe all characteristics of the data proposed for every alternative solution.

C.9.4. DELIVERY REQUIREMENTS

All data deliveries shall meet the following requirements as described in this section. Each Task Order RFP may contain more specific delivery requirements.

1. Provide Level 0, 1 and/or 2 data as specified in IDIQ SOW section C.9.3. Each Task Order RFP will specify more refined data level delivery requirements.
2. Contractors shall deliver data to NOAA on a regular update cadence suitable for operational data assimilation, rather than as a continuous stream.
3. The Contractor shall provide data deliveries at a nominal cadence of no less than once every 3 hours, unless specified in a Task Order.
4. Each delivery shall include the most recent 24 hours of available data, updated to reflect the latest orbit determination and processing results.
5. The end time of the delivered data window shall lag real time by no more than the delivery latency, which may be on the order of tens of minutes to an hour, as appropriate to the Contractor's orbit determination and processing methodology.
6. NOAA acknowledges that OD-derived parameters may have an inherent temporal lag due to the estimation process weighting over multiple orbits.
7. Periods affected by maneuvers, thrusting, or non-nominal operations shall be clearly flagged. NOAA may exclude these periods from operational density estimation or assimilation.
8. Data shall conform to NESDIS and NCCF ingest standards.
9. Metadata shall describe:
    - Time systems and reference frames,
    - OD, drag, and/or density derivation methodologies,
    - Quality flags and known limitations.
10. Data formats may include NetCDF, HDF5, or other NOAA-approved formats.

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C.9.5. USE OF DATA

NOAA may use the delivered data to derive or ingest in-situ or orbit-effective thermospheric neutral density using Government-developed, Contractor-furnished, or jointly developed methodologies. These data and products may be used to generate near-real-time and retrospective geophysical products, assimilate density estimates into operational thermosphere and space weather models, and support SSA, orbit prediction, and research applications.

NOAA may consider alternative, more restrictive sharing rights for Thermospheric Neutral Density data. Orbit-effective thermospheric neutral density products derived from Contractor-provided data, may be treated as derived or value-added products in accordance with the data rights provisions as specified in a Task Order RFP.

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FC7: Active Sensors Requirements

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C.10. Functional Category 7: ACTIVE SENSORS DATA IDIQ REQUIREMENTS

C.10.1. INTRODUCTION

The National Oceanic and Atmospheric Administration (NOAA) is dedicated to understanding and predicting changes in weather, oceans, and coastal environments. NOAA shares this knowledge widely and manages coastal and marine ecosystems and resources as the Nation's authoritative environmental intelligence agency.

NOAA mission requirement is the acquisition of accurate, global, and temporally frequent measurements of ocean/sea surface properties such as wind speed and direction, surface and wave heights and precipitation. Such observations are essential to generate a variety of sea-surface products for marine and weather forecasting and weather and ocean monitoring. In addition, these observations provide actionable intelligence for sea-ice characterization, soil moisture estimation, flood and inundation monitoring, disaster management, and assimilation into NWP models for improved forecast skill. NOAA seeks to procure data from Active Sensors such as Radar Altimetry, K-Band radar, Synthetic Aperture Radar (SAR), and Scatterometry.

Radar altimeters measure the two-wave travel time of a radar pulse between the satellite antenna and the Earth's surface at the nadir or the spacecraft. Altimetry missions include complementary instruments to ensure precise location and altitude knowledge at the time of the measurement. NOAA primarily uses altimetry measurements to observe sea surface height, wave heights, and wind speeds at km footprint scales.

K-Band (Ku, K, Ka) radars offer a novel technique to actively measure precipitation from a space-based platform. K-Band radars offer insight into vertical profiles of precipitation, precipitation intensity and precipitation distribution which is essential for accurately monitoring and forecasting severe weather events. This technology offers a vast spatial-temporal improvement over traditional land-based radar measurement of precipitation. NOAA will utilize these measurements for weather monitoring and forecasting applications.

SAR imagery is utilized to generate a variety of surface products along land, coastal, and ocean regions. It provides an invaluable resource for environmental monitoring, marine surveillance, disaster management, sea-ice characterization, resource management and mapping. SAR data is most useful for surface wind products, sea ice products, and land/sea surface property products. SAR Imagery can be collected in all weather conditions, both day and night.

Scatterometers measure a backscattered signal to derive wind speed and direction from the roughness of the sea. The sensor may use radio or microwave bands to determine the

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normalized radar cross section of a surface. Scatterometry data is especially useful in estimating ocean/sea surface wind speed and direction.

C.10.2. SCOPE

NOAA is soliciting commercial, near-real-time satellite-based observation data from Active Sensors (Altimetry, K-Band radar, SAR, Scatterometry) to support a variety of environmental monitoring and weather forecasting applications. NOAA will process these measurements into geophysical products and incorporate them into downstream applications and archive both the raw data and the derived datasets for long-term access and scientific research.

C.10.3. DATA REQUIREMENTS

All delivered data, including associated metadata and ancillary files, shall conform to NOAA data standards or be fully compatible with NOAA operational data systems, formats, and protocols, to ensure seamless integration, processing, and archival within NOAA's enterprise systems.

C.10.3.1. Definitions

The following definitions are provided in the context of all Active Sensor measurements:

12. **Altimetry:** An active microwave remote sensing system that transmits radar or laser pulses from a satellite towards the Earth's surface and measures the time it takes for the signals to bounce back. This technique allows scientists to determine the vertical distance between the satellite and the surface.

13. **Antenna Gain**: The measure of how well an antenna directs radio frequency energy in a specific direction, affecting both transmitted and received signal strength.

14. **K-Band (Ku, K, Ka) Radar:** An active microwave remote sensing system that transmits radar pulses from a satellite towards the Earth's surface and measures the return echoes to determine precipitation properties. This IDIQ definition of K-Band includes Ku, K, Ka bands.

15. **Latency**: The time interval between the observation of the data and the receipt of the data into the NESDIS Common Cloud Framework (NCCF).

16. **Level 0**: Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed. This may include raw intermediate frequency (IF) samples or complex baseband representations without calibration, geolocation, or additional metadata.

17. **Level 1**: Instrument-processed, calibrated data at full resolution, time-referenced, and annotated with ancillary information. This may include derived data with associated metadata and basic geolocation.

18. **Level 2**: Geophysical retrieval-ready observables derived from calibrated Level 1 data. These observable data shall be suitable for NOAA environmental retrievals for scientific and operational use.

19. **Scatterometry:** An active microwave remote sensing technique that transmits microwave pulses and analyzes the backscattered signal to measure the normalized radar cross-section and infer geophysical parameters such as ocean wind speed and direction, sea ice properties, and land surface characteristics like soil moisture. Scatterometry leverages the principle that the roughness and dielectric constant of the surface affect the amount of energy scattered back to the sensor.

20. **Synthetic Aperture Radar (SAR):** An active microwave remote sensing system that transmits radar pulses and measures the amplitude and phase of the backscattered energy from the Earth's surface to generate high-resolution, all-weather imagery. SAR can measure the normalized radar cross-section to infer geophysical parameters including ocean surface wave patterns, sea ice extent, and soil moisture, utilizing the sensitivity of the microwave signal to surface roughness and dielectric properties.

C.10.3.2 Active Sensor Data Requirements

The contractor shall deliver data from Active Sensor systems. The specific Level 0, Level 1 and Level 2 data requirements will be described in each Task Order RFP.

1. Observations shall be delivered at one or more of the following levels:

   a. Level 0 data including:
      - iii. Full resolution data.
      - iv. Compressed data.

   b. Level 1 A and B products including:
      - i. Calibrated data. Observations will include all data necessary to achieve fully calibrated data. Data must be radiometrically calibrated to correct for system biases, antenna patterns, and platform effects.

   c. Level 2 products including (but not limited to):
      - i. Ocean/Sea Surface Wind and Wind Vector products
      - ii. Ocean/Sea Surface Height products (Altimetry)
      - iii. Ancillary and complementary products

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iv. Precipitation estimates (K-Band Radar)

d. Provide the following related data for use in NOAA's analysis team:
   i. Quality flags generated in processing pertaining to all levels of provided data and relevant documentation. Quality flags will include signal quality, land/ocean discrimination, ice flags, and navigation or receiver errors.
   ii. All data used for calibration/validation and product development.
   iii. Data readers and format descriptions (i.e., reader software source, data dictionaries, etc.) necessary for processing and analysis.

C.10.4. OBSERVATION REQUIREMENTS

2. NOAA will utilize active sensor measurements for specific applications.

   a. Active sensor measurements observations for the collection of ocean/sea surface wind observations (SAR, Scatterometry) shall meet or exceed the parameters summarized in Table 1.

   b. Active sensor measurements observations for the collection of sea surface height observations (SAR, Altimetry) shall meet or exceed the parameters summarized in Table 2.

   c. SAR imagery requirements shall meet or exceed the parameters summarized in Table 3.

   d. K-Band measurements for precipitation shall include measurements from within one or more of the following frequency bands: Ku, K, Ka.

3. Each Task Order RFP will contain more specific observation requirements to include, but not limited to, observation quantities, frequency band coverage, swath, resolution, sensitivity, polarization, accuracy, latency, refresh and coverage.

Table 1. Active Sensor measurement requirements for ocean/sea surface wind observations
(Applicable to SAR and Scatterometry technologies)

Performance Level	**Threshold Requirement**
Horizontal Resolution	25 km
Accuracy (Direction for wind speed > 5 m/s) (only applies to Scatterometry)	20 degrees
Accuracy (Speed)	2 m/s or 10% of the measurement
Latency	180 minutes

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Table 2. Active Sensor measurement requirements for sea surface height observations
(Applicable to SAR and Altimetry technologies)

System Capability	Threshold Requirement
Horizontal Resolution	30 km
Accuracy	3.4 cm
Latency	24 hours

Table 3. SAR Imagery requirements
(Applicable to SAR technologies)

System Capability	Threshold Requirement
Horizontal Cell Size	250 m
Latency	8 hours

4. The daily count of compliant observations and/or the minimum aggregate data availability from a constellation will be specified in each Task Order RFP.
5. Geographic distribution of all data shall be worldwide or as specified in each Task Order RFP. Contracts shall have the ability to provide data over regional areas as specified in a Task Order RFP.
6. Contractors shall have tasking ability of satellite(s) to provide higher quantities of system observations during certain events (such as tropical cyclones), as specified in a Task Order RFP.

C.10.5. DELIVERY REQUIREMENTS

All Active Sensor measurements shall meet the following requirements as described in this section. Each Task Order RFP may contain more specific delivery requirements.

3. At a minimum, the Contractor must have the capability to provide continuous data for one satellite. One satellite of data represents the minimum ordering quantity for Task Orders.
4. Metadata shall include information for each satellite/instrument configuration.
5. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. Technical documentation shall include, at a minimum, the following:

a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.
b. Documentation to support quality control and calibration/validation activities.
c. Antenna pattern designs and gain patterns, if appropriate.
d. User guides and detailed data format and data specification documents.
e. Description of instrument(s) used in the data acquisition and their sampling characteristics.
f. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

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Additional SBEM Requirements

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C.11. ADDITIONAL SPACE-BASED ENVIRONMENTAL MONITORING (SBEM) IDIQ REQUIREMENTS

C.11.1. INTRODUCTION

The National Oceanic and Atmospheric Administration (NOAA) understands the importance of incorporating additional data types that could help advance NOAA's environmental applications. In addition to the Functional Categories listed in this IDIQ, NOAA may consider additional Space-Based Environmental Monitoring (SBEM) data types that could meet its mission needs and, as such, may on-board Contractors for these additional data types.

C.11.2. SCOPE

NOAA is soliciting commercial, near-real-time satellite-based observation data for additional data types across all orbits to support a variety of environmental monitoring and weather forecasting applications. NOAA may consider additional capabilities from space-based platforms to support additional environmental product areas. NOAA will process these measurements into geophysical products and incorporate them into downstream applications and archive both the raw data and the derived datasets for long-term access and scientific research.

C.11.3. DATA AND DATA DELIVERY REQUIREMENTS

1. At a minimum, the Contractor must have the capability to provide continuous data for one satellite. Exact minimum ordering quantities will be determined in future Task Order RFPs.

2. The Contractor shall deliver Earth-relevant data products obtained from commercial small satellites, associated metadata, and ancillary information. All delivered data, including associated metadata and ancillary files, shall conform to NOAA data standards or be fully compatible with NOAA operational data systems, formats, and protocols, to ensure seamless integration, processing, and archival within NOAA's enterprise systems.

3. The Contractor shall, at a minimum, deliver applicable Level 0 and Level 1 data.

4. Metadata shall include information for each satellite/instrument configuration.

5. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. Technical documentation shall include, at a minimum, the following:

a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.
b. Documentation to support quality control and calibration/validation activities.
c. Antenna pattern designs and gain patterns, if appropriate.
d. User guides and detailed data format and data specification documents.
e. Description of instrument(s) used in the data acquisition and their sampling characteristics.
f. Pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.

6. NOAA will describe more specific information on data, data formats and data delivery requirements in future Task Order RFPs.

C.11.4. OBSERVATION REQUIREMENTS

7. The government may require data types from a space-based platform that meets one or more of the Geophysical products as listed in Table 1. Each Task Order RFP will contain more specific observation requirements to include, but not limited to, accuracy, coverage, data formats, data types, frequency band coverage, latency, observation quantities, refresh, resolution, sensitivity, and swath.

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Table 1. Geophysical Product Categories
(reference: [The NESDIS Level Requirements (NESDIS-REQ-1001.1](https://)))

Geophysical

Atmosphere	Cryosphere	Land & Surface Hydrology	Oceans, Freshwater and Coasts	Space
Atmospheric Composition and Air Quality	Lake and Sea Ice	Fires	Biology and Biogeochemistry	Heliosphere
Atmospheric Temperature	Snow and Glaciers	Flood	Surface Height	Ionosphere
Atmospheric Water Vapor		Surface Moisture	Topography and Bathymetry	Magnetosphere
Clouds		Surface Temperature	Water Pollution	Solar
Lightning		Vegetation	Water Temperature and Salinity	
Precipitation				
Radiation Budget				
Tropical Cyclone Characteristics				
Volcanic Eruption Characteristics				
Winds				